Arab Open University- Egypt



Faculty of Computer Studies
Cyber Security Department

Startup Company Infrastructure

Roaa Ramadan Ahmed
22512817

TM471: Final Year Project, May 2026

Supervisor:
Dr. Marwa Khashaba
Dr. Ibrahim Elhasoni

2025/2026

Abstract

Many startup organizations suffer from an unstructured and insecure network environment, where the absence of centralized management and proper security controls results in operational inefficiencies and increased vulnerability to cyber threats.

This project presents a comprehensive design and implementation plan for building a secure, scalable, and centrally managed IT infrastructure for a startup company using Windows Server technologies, advanced network simulation through GNS3, and enterprise-grade protection provided by a FortiGate firewall.

GNS3 is used to simulate the complete network, including routers, switches, servers, VLANs, and security layers. The FortiGate firewall ensures secure segmentation, strong traffic filtering, intrusion prevention, and centralized monitoring across the infrastructure environment.

Acknowledgments

Firstly, I just say Alhamdulillah for everything and thank Allah for guiding me to undertake this.

I would like to express my deep appreciation to all individuals who supported the development of this project. My sincere gratitude goes to my supervisor Dr. Marwa Khashaba & Dr. Ibrahim Elhasoni for providing valuable guidance, constructive feedback, and continuous encouragement. Their expertise played a major role in shaping the structure and decision-making throughout the implementation stages.

Finally, I express heartfelt gratitude to my family and friends for their endless support, patience, and encouragement. Their belief in my abilities has always been a source of strength throughout my academic journey.

Contents

Abstract ii
Acknowledgments iii
Chapter 1 9
1.1 Background 10
1.2 Problem Statement 10
1.3 Aims and Objectives 10
1.3.1 Aims 10
1.3.2 Objectives 11
1.4 Scope 11
1.5 Deliverables 11
1.6 Target Users 12
1.7 Suggested Solution 12
1.8 Summary 12
1.9 Project TimeLine 13
Chapter 2 14
2.1 Introduction 15
2.2 Background 15
2.3 Relevant Technologies 17
2.3.1 Microsoft Windows Server 18
2.3.2 GNS3 Network Simulation [4] 19
2.3.3 FortiGate Firewall [3] 19
2.4 Related Work and Existing Systems 20
2.5 Comparison 22
2.6 Additional Literature Insights and Discussion 22
2.7 Summary 23
Chapter 3 24
3.1 Introduction 25
3.2 Functional Requirements 25
3.3 Non-Functional Requirements 26
3.4 Network Design Analysis [3] 26
3.5 Software and Hardware Requirements 27
3.5.1 Software Requirements 27
3.5.2 Hardware Requirements 27
3.6 Diagrams 28
3.7 System Architecture Summary 35
3.8 Code of Ethics 35
3.9 Summary 35
Chapter 4 36
4.1 Overview 37
4.2 Design 37
4.3 Implementation 38
4.3.1 Used Protocols 38
4.3.2 VLAN Segmentation 39
4.3.3 IOU1 40
4.3.4 IOU2 43
4.3.5 R1 45
4.3.6 FortiGate Firewall 46
4.3.7 DHCP and IP Address Management 50

4.3.8 Windows Server and Active Directory 51
4.4 Dashboard Design and Implementation 54
4.4.1 Dashboard Frontend Design 54
4.4.2 Technologies Used 55
4.4.3index.html 56
4.4.4 app.js 57
4.4.5 network_monitor.py 59
4.4.6 Real-Time JSON Data Generation 60
4.4.7 Dashboard Features Summary 61
4.5 Tools and Libraries 62
4.6 System Testing and Validation 63
4.6.1 Connectivity Testing 63
4.6.2 VLAN and Inter-VLAN Routing Validation 63
4.6.3 HSRP Redundancy Testing 64
4.6.4 Rapid Spanning Tree Protocol (RSTP) Testing 65
4.6.5 EtherChannel Validation 66
4.6.6 Firewall Connectivity Testing 67
4.6.7 DHCP Service Validation 68
4.6.8 Dashboard Monitoring Validation 68
4.7 Network Security and Protection 71
4.8 Inclusion, Diversity, and Ethical Compliance 71
4.9 Resilience and Fault Management 71
Summary 72
Chapter 5 73
5.1 Overview 74
5.2 Experimental Results 74
5.3 Discussion 75
5.4 Goals Achieved 75
5.5 Strengths and Limitations. 76
5.6 Future Work 76
Summary 77
Chapter 6 78
6. Final Conclusion 79
References 81

List of Figures

Figure 1: Gantt Chart 13
Figure 2: Hierarchical Network Design 16
Figure 3: Use Case Diagram 28
Figure 4: Sequence Diagram 30
Figure 5: Activity Diagram 33
Figure 6: Data Flow Diagram 34
Figure 7: Full Network Topology 37
Figure 8:VLAN Configuration 39
Figure 9:IOU1 40
Figure 10:Spanning-tree configuration 40
Figure 11:EtherChannel configuration 41
Figure 12:Trunking 41
Figure 13:SVI 42
Figure 14: static route 42
Figure 15:IOU2 43
Figure 16:STP Configuration 43
Figure 17:Gateway Redundancy using HSRP 43
Figure 18:Link Aggregation using LACP 44
Figure 19:VLAN Trunk Configuration 44
Figure 20:DHCP Configuration 44
Figure 21:Default Routing Configuration 45
Figure 22:Transit Routing Infrastructure 45
Figure 23:Upstream Default Routing 45
Figure 24:Floating Static Route Failover 46
Figure 25:FortiGate Interface Configuration 46
Figure 26:Internal VLAN Routing 47
Figure 27:Routing Table 47
Figure 28:IT & Manager Policy 48
Figure 29:HR & Sales Policy 48
Figure 30:Finance & Block Policy 49
Figure 31:Firewall Security Policies 49
Figure 32:DHCP 50
Figure 33:DHCP Configuration 50
Figure 34:WinServer2022 51
Figure 35:Active Directory User Management 51
Figure 36:DNS Service Configuration 52
Figure 37:GPO Configuration 52
Figure 38:Password Security Policy 53
Figure 39:NetWatch Login Interface 54
Figure 40:NetWatch Real-Time Monitoring Dashboard 55
Figure 41:Dynamic metrics 56
Figure 42:Device Monitoring 56
Figure 43:Fault Detection 57
Figure 44:VLAN Definition 57
Figure 45:Dynamic JSON 57
Figure 46:Fault Analyzer 58
Figure 47:Auto Refresh 58
Figure 48:Login Authentication 58

Figure 49:Device Definitions....59
Figure 50:ICMP Ping Monitoring....59
Figure 51:Alert Generation....60
Figure 52:JSON Generation....60
Figure 53:ping form IOU1 to server....63
Figure 54:Ping from vlan60 to vlan20....63
Figure 55:Routing....64
Figure 56:HSRP Active and Standby....64
Figure 57:RSTP Root Bridge and Forwarding State Verification....65
Figure 58:EtherChannel and LACP Validation....66
Figure 59:Internet Connectivity....67
Figure 60:DHCP Pool and Client Address....68
Figure 61:Dashboard....69
Figure 62:Json....69
Figure 63:Network scan....70

List of Tables

Table 1: Scheduling Table 13
Table 2: Comparison among different approaches 22
Table 3: Non-Functional Requirement 26
Table 4:VLANs 39
Table 5:Services 53
Table 6:Technologies used 55
Table 7:Ping 59
Table 8:Dashboard summary 61
Table 9:Testing 70
Table 10:Summary of network testing results 74

Chapter 1
Introduction

1.1 Background

A startup company is typically established with a small team, limited resources, and a high dependency on technology for communication, collaboration, and service delivery. As the company grows, the number of employees, devices, applications, and internal systems increases significantly.

A startup requires:

- Centralized user authentication
- Secure access to shared resources
- Separation between departments
- Strong firewall protection
- Fast troubleshooting
- Scalable network architecture
- Efficient device and user management

1.2 Problem Statement

Many medium-sized enterprises struggle with decentralized network management and inadequate security measures, leading to data breaches, inefficient resource allocation, and high administrative overhead. Specifically, for a company operating across multiple floors without a structured hierarchy, managing individual devices and securing inter-departmental communication is prone to errors.

The lack of planning creates critical weaknesses:

- No centralized user login system
- Inconsistent access control
- Manual configuration of devices
- Weak or nonexistent security policies
- Flat network architecture with zero segmentation
- Vulnerability to cyberattacks
- Lack of real monitoring and logging

1.3 Aims and Objectives

1.3.1 Aims

The main aim of this project is to design and implement a complete IT infrastructure for a startup company using Windows Server, GNS3 simulation, and FortiGate firewall security to ensure centralized management, scalability, and data protection.

1.3.2 Objectives

Technical: To deploy and configure Windows Server 2022 for Active Directory Domain Services (AD DS), DNS, and DHCP

Security: To implement FortiGate Firewall policies to control traffic between the internal network (LAN) and the external network (WAN)

Social: Improve collaboration within the company. Provide organized access to shared data

Ethical: Ensure confidentiality of sensitive business information

Scientific: Apply standards of IT infrastructure design used in corporate environments

---

1.4 Scope

- Full Windows Server configuration
- DNS and DHCP service deployment
- VLAN design and network segmentation
- GNS3 simulation topology
- Firewall rule design using FortiGate
- File server implementation
- Policy enforcement through GPO

1.5 Deliverables

- **Infrastructure & Network Design**: Logical and physical design of the startup network, including VLANs, IP scheme, and placement of core services (AD DS, DNS, DHCP, File Server, FortiGate).
- **GNS3 Simulation Topology**: A working GNS3 project that emulates the full network to test connectivity, routing, and basic security policies before real deployment.
- **Windows Server & FortiGate Configuration Summary**: Documented setup steps and key settings for Windows Server 2022 (domain, DNS, DHCP, GPOs) and FortiGate firewall policies used to secure and control traffic.
- **Testing Overview**: A brief validation of main functions such as user login, file sharing, inter-VLAN communication, and Internet access control.

1.6 Target Users

- Regular Employees
- HR and Finance Personnel
- IT and System Administrators
- Management and Decision Makers
- Future Students and Researchers

1.7 Suggested Solution

To address the challenges identified in the existing unstructured and insecure network environment, this project proposes a comprehensive and scalable IT infrastructure solution based on industry-standard technologies. The suggested solution integrates **Windows Server 2022**, **GNS3 network simulation**, and a **FortiGate next-generation firewall** to create a secure, centrally managed, and highly organized enterprise environment suitable for a growing startup, and how businesses can reduce downtime, enhance data protection, and streamline IT administration.

1.8 Summary

This chapter introduced the project background, problem statement, and objectives. It highlighted the need for a secure and centrally managed IT infrastructure for a startup company. The chapter also outlined the proposed solution, project scope, methodology, and significance, providing a clear foundation for the rest of the study.

1.9 Project TimeLine

1.9.1 Scheduling Table

Table 1: Scheduling Table

	Start Date	End Date	DESCRIPTION
1	1-10	9-10	Find the project idea
2	11-10	14-10	Proposal
3	15-10	20-10	chapter 1
4	23-10	30-10	chapter 2
5	2-11	12-11	chapter 3
6	15-11	25-11	End of report 1
7	28-11	8-12	Presentation 1

1.9.2 Gantt Chart

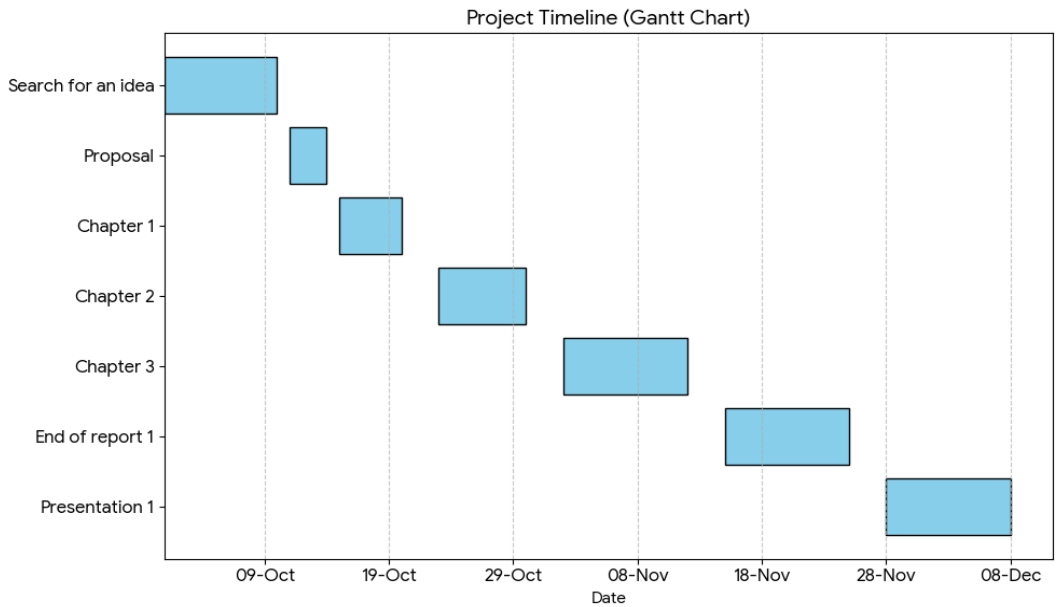


Figure 1: Gantt Chart

Chapter 2
Literature Review

2.1 Introduction

A well-designed IT infrastructure is crucial for modern organizations, especially startups that face rapid growth, rising operational demands, and an ever-increasing dependence on digital systems. As businesses expand, the number of employees, devices, services, and applications grows exponentially, making traditional ad-hoc networks insufficient, insecure, and difficult to manage. Therefore, academic and industry research highlights the importance of adopting structured, scalable, and centrally managed IT environments.

This chapter provides a detailed literature review of the foundational technologies and methodologies that support the design of a secure corporate infrastructure for a startup company. The section discusses the conceptual background of enterprise networking, examines the role of Microsoft Windows Server in identity and access management, evaluates the relevance of network simulation tools such as GNS3, and analyzes the security contributions of FortiGate next-generation firewalls. Additionally, the chapter explores related academic works, compares various approaches used in previous studies, and highlights how the proposed system overcomes existing limitations.

The literature consistently emphasizes that startups—despite their smaller size—require infrastructure similar to large enterprises in terms of reliability, security, and centralization. This is because the transition from a few employees to a multi-department organization can happen rapidly, and failing to establish a proper technological foundation early can lead to inefficient processes, cybersecurity vulnerabilities, and high administrative costs. As such, the adoption of Active Directory, firewall solutions, structured network segmentation, and virtualized testing environments provides both immediate operational benefits and long-term stability for the company.

2.2 Background

IT infrastructure in a business environment integrates multiple layers of technology, including networking, server administration, security tools, and organizational processes. Theoretical models such as the **Hierarchical Network Design Model** and **Centralized Identity Management** play a fundamental role in shaping modern infrastructures.

Modern business networks must accommodate increasing numbers of users, devices, and services while maintaining high levels of performance and security. Research in computer networking and IT infrastructure design highlights several concepts essential to building reliable corporate environments, many of which are particularly relevant to rapidly growing startups.

1. Hierarchical Network Design Model [2]

The hierarchical model divides the network into three logical layers:

1. **Core Layer**: Responsible for high-speed data transmission across the network.
2. **Distribution Layer**: Controls policies, routing, and management.
3. **Access Layer**: Connects end devices such as employee PCs, printers, and wireless access points.

Literature from Cisco and other networking authorities argues that this layered approach is essential for:

- Reducing congestion and collision domains
- Increasing network reliability
- Facilitating easier troubleshooting
- Enhancing security by enabling better segmentation

Startups that skip hierarchical design often suffer from flat network architectures, leading to broadcast storms, performance reduction, and weak internal security.

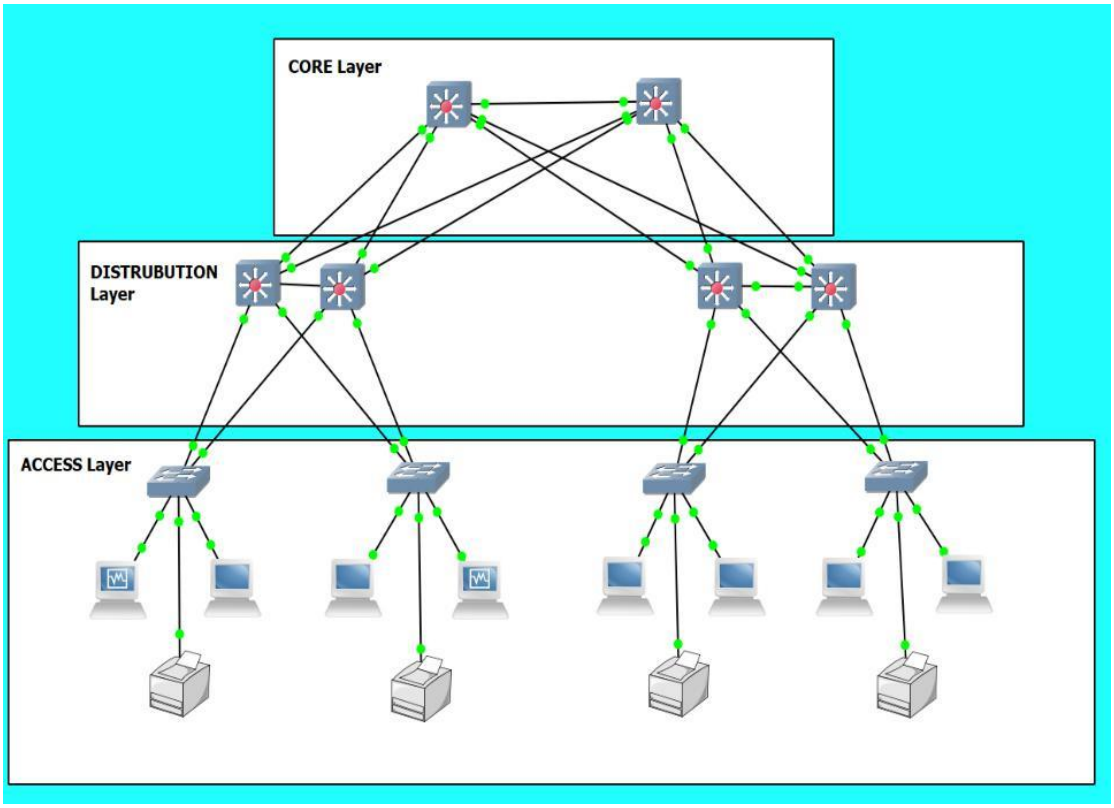


Figure 2: Hierarchical Network Design

2. Centralized Identity and Access Management

Identity management is a cornerstone of corporate environments, relying on authentication and authorization frameworks to control access. **Active Directory Domain Services (AD DS)** is the most widely implemented system for centralized authentication. Research shows that AD DS provides:

- A unified login system for employees
- A central location for managing devices, users, and permissions
- Group policies that enforce consistent configuration
- A secure authentication mechanism using Kerberos

Startups without centralized identity management often face operational chaos, including inconsistent permissions, insecure data sharing practices, and difficult troubleshooting processes. Implementing AD DS early ensures long-term scalability and consistent organizational structure.

3. Network Security and Segmentation

Another core theoretical concept is **network segmentation**, which divides the network into isolated zones (VLANs). Studies show segmentation significantly enhances security by:

- Limiting lateral movement attacks
- Containing malware outbreaks
- Reducing broadcast traffic
- Separating sensitive departments such as HR and Finance

Firewalls and ACLs enforce rules between these segments.

4. Virtualization in IT Infrastructure

Virtualization—through tools like VMware or GNS3—enables organizations to replicate real IT environments without purchasing physical hardware. Modern literature highlights that virtualization is particularly beneficial for startups because it:

- Lowers initial infrastructure costs
- Enables rapid testing and prototyping
- Allows isolated experiments without harming production systems
- Supports scalability and flexibility

GNS3 adds another layer of realism by providing router, switch, and firewall emulation, which is essential for validating architectural decisions before deployment.

2.3 Relevant Technologies

The proposed system integrates three core technologies:
Windows Server 2022, **GNS3 Network Simulator**, and **FortiGate Next-Generation Firewall**.
Each technology plays a unique role in forming a secure and efficient startup infrastructure.

2.3.1 Microsoft Windows Server

Key Roles and Features from Literature

1. **Active Directory Domain Services (AD DS)** [6]
   AD DS is responsible for authentication, authorization, and directory services. It ensures users log into the domain securely, and administrators can manage accounts centrally. Literature highlights AD DS as the foundation for corporate IT environments, enabling:

- Group-based policy enforcement
- Secure authentication through Kerberos
- Hierarchical organizational unit (OU) structure
- Scalable identity management

2. **Domain Name System (DNS)**
   DNS in Windows Server is integrated with AD DS, providing automatic updates and secure zone transfers. Research shows that internal DNS is essential for service discovery, domain controller location, and resource management. [7]

3. **Dynamic Host Configuration Protocol (DHCP)**
   DHCP automates IP assignment, reducing administrative burden. The protocol ensures. [5]

- Correct gateway and DNS configuration
- Automated updates
- Reduced risk of IP conflicts

4. **Group Policy Objects (GPOs)**
   GPOs allow IT administrators to apply organization-wide settings to computers and users. Literature emphasizes their importance for:

- Applying security configurations
- Restricting unauthorized software
- Enforcing password policies
- Mapping network drives

5. **File and Storage Services**
   Windows Server supports NTFS permissions and SMB protocol for managing shared folders, ensuring controlled access within departments.

Relevance to the Proposed Project

Windows Server 2022 forms the backbone of the infrastructure. It enables:

- Centralized management

- Defined user roles
- Secure authentication
- Controlled access to sensitive data

For a startup, this prevents the chaos of unmanaged workstations, inconsistent configurations, and weak internal security.

2.3.2 GNS3 Network Simulation [4]

GNS3 is a network emulation platform capable of running real operating systems and network appliances. Unlike simple simulators (e.g., Cisco Packet Tracer), GNS3 allows integration of actual Windows Server images, FortiGate appliances, and router/switch firmware.

Key Advantages in Literature

- **Realistic Emulation**: Supports real Cisco IOS images, real firewalls, and Windows Server virtual machines.
- **Cost Efficiency**: No need for physical network hardware during the design phase.
- **Safe Testing Environment**: Administrators can experiment with misconfigurations, attacks, routing loops, and firewall rules.
- **Complex Network Topologies**: Supports VLANs, routing, multi-layer switching, and NAT testing.
- **Scalability**: Startups can simulate medium-to-large infrastructures before committing resources.

Researchers note that simulation reduces time, cost, and deployment risks.

Relevance to the Project

GNS3 is used to simulate:

- VLAN segmentation
- Routing between subnets
- Windows Server behavior
- FortiGate firewall placement
- DHCP, DNS, and AD DS functionality

By validating everything virtually, the company ensures zero downtime and reduces implementation costs

---

2.3.3 FortiGate Firewall [3]

FortiGate is a next-generation firewall (NGFW) known for its high performance and advanced threat protection. Literature places FortiGate among the top security

appliances globally due to its combination of network security, application control, and threat intelligence integration.

Key Features from Literature

1. **Deep Packet Inspection (DPI)**
   FortiGate examines packet contents, blocking threats that pass through traditional firewalls.
2. **Intrusion Prevention System (IPS)**
   IPS identifies anomalous or malicious activity using real-time threat signatures.
3. **Application Control**
   Allows administrators to block unauthorized applications such as torrents or remote access tools.
4. **Web Filtering**
   Prevents access to harmful or inappropriate websites.
5. **SSL Inspection**
   Decrypts encrypted traffic for scanning, enhancing visibility.
6. **Virtual Private Network (VPN)**
   Supports secure remote connections using IPsec and SSL VPN.
7. **Traffic Shaping and Bandwidth Control**
   Ensures fair distribution of network resources.
8. **FortiView Monitoring Dashboard**
   Provides real-time visibility into network traffic, threats, and user behavior.

Relevance to the Proposed Project

FortiGate acts as the perimeter and internal security guardian. It enforces:

- Segmentation between departments (HR, Finance, IT, Employees)
- Controlled Internet access
- Protection of Windows Server resources
- Secure routing and VPN capabilities
- Deep inspection of traffic

For a startup handling sensitive data, FortiGate ensures protection against external and internal threats.

2.4 Related Work and Existing Systems

A review of existing literature and previous implementations reveals that network infrastructure projects have been approached differently depending on the tools, objectives, and constraints of the environment.

o Design of Secure Campus Network using Cisco Packet Tracer (Al-Bahadili, 2020) [1]

- **Contribution:** Designed a network for a university campus focusing on VLANs and ACLs.

- **Methodology:** Used Cisco Packet Tracer simulation.

- **Strengths:** Cost-effective and good for logical design visualization.

- **Limitations**: Packet Tracer is a simulator, not an emulator; it lacks the full features of a real operating system like Windows Server 2022 or FortiGate.

○ Implementation of Active Directory on Windows Server 2016 (Smith & Jones, 2019) [9]

- **Contribution**: Focused on deploying AD DS to manage user authentication in a small business.

- **Methodology:** Deployed on physical hardware.

- Strengths: Real-world performance analysis.

- **Limitations:** Used an older version of Windows Server (2016) and lacked a dedicated hardware firewall implementation, relying only on software firewalls.

○ Flat Network Topology Performance Analysis (Chen et al., 2021)

- **Contribution:** Analyzed the performance of small office networks using a flat (non-hierarchical) topology.

- **Methodology:** Simulation of a single-layer network.

- **Strengths:** Simple to implement for very small sites.

- **Limitations:** Showed significant broadcast storms and management difficulties as the number of nodes increased, proving it unsuitable for multi-floor companies.

2.5 Comparison

Table 2: Comparison among different approaches

Author/System	Approach	Strength	Limitation
Al-Bahadili (2020)	Packet Tracer Simulation	Easy to visualize logical topology	Limited commands; Cannot run real Windows Server OS
Smith & Jones (2019)	Physical Windows Server 2016	Real hardware performance	Outdated OS; High cost for hardware; No dedicated Firewall appliance
Proposed System	GNS3 Emulation + Win Server 2022	Runs actual OS images; Modern Hierarchical Design; FortiGate Security	Requires high computing resources for emulation

2.6 Additional Literature Insights and Discussion

Several additional theoretical and practical considerations further justify the relevance and necessity of the proposed infrastructure. Modern IT environments especially those supporting startups must account for not only performance and security, but also adaptability, compliance, operational efficiency, and future technological integration. Thus, deeper exploration of virtualization, network management frameworks, and cybersecurity methodologies reveals why the chosen stack (Windows Server 2022, GNS3, and FortiGate) represents an optimal, research-backed solution.

One of the most important developments in recent IT literature is the transition from traditional hardware-centric infrastructure toward virtualized and software-defined environments. Virtualization reduces dependency on physical servers, lowers operational costs, and allows companies to test and deploy new systems at significantly faster rates. Studies on virtualization efficiency demonstrate that using tools like VMware or Hyper-V reduces downtime and increases fault tolerance by enabling snapshots, cloning, and rapid disaster recovery. GNS3, while primarily a network simulation platform, aligns with these principles by allowing engineers to simulate routers, switches, and firewalls without requiring physical devices. Academic research repeatedly confirms that simulation-based design minimizes

implementation errors, as networks can be fully tested before deployment in real environments.

Another emerging trend in IT infrastructure literature is the importance of centralized monitoring and logging. Modern organizations must ensure visibility into network behavior to meet cybersecurity standards and regulatory policies. Tools such as FortiView and Windows Server auditing features provide real-time insights into authentication attempts, resource usage, system performance, and security events.

Additionally, academic discussions emphasize scalability as a critical success factor in infrastructure design. Startups often transition from small teams to medium-sized enterprises within a few years, requiring networks to expand seamlessly without redesigning the entire architecture. The hierarchical model used in this project is directly supported by literature as the most scalable network design. Each layer—core, distribution, and access—can grow independently, allowing new departments, servers, and firewalls to be integrated with minimal disruption. Windows Server further enhances scalability, as its Active Directory can be extended with additional domain controllers, child domains, and organizational units as needed.

2.7 Summary

This chapter reviewed the relevant literature and technologies related to enterprise network infrastructure. It discussed Windows Server, GNS3 simulation, and FortiGate firewalls, and analyzed related work to justify the proposed solution. The comparison demonstrated the advantages of using modern, enterprise-grade technologies for startup environments.



Chapter 3
Requirements and Analysis

3.1 Introduction

This chapter presents a detailed analysis of all the requirements necessary for designing and implementing a complete and secure startup company IT infrastructure using Microsoft Windows Server, GNS3 simulation, and a FortiGate firewall. Requirement analysis is considered one of the most critical phases in infrastructure design because it lays the foundation upon which the architecture, configuration, and implementation will be based.

3.2 Functional Requirements

1. **Active Directory (AD DS)**

   Active Directory Domain Services is the core of identity and access management in the startup.

2. **DNS**

   DNS plays a critical role in resolving hostnames within the internal network.

3. **DHCP**

   DHCP automates the assignment of IP addresses, reducing administrative effort.

4. **Group Policy (GPO)**

   GPOs must enforce password complexity, desktop restrictions, access rules, software policies, USB limitations, and firewall configurations. Policies must be linked to department-specific OUs and verified for accurate application.

5. **File Server**

File servers host shared data for each department and company-wide public folders. Access is controlled using NTFS permissions and security groups so that only authorized users can open sensitive HR or Finance documents, while supervisors may have extended privileges.

6. **GNS3 Simulation and FortiGate**

   The complete design is simulated in GNS3. The simulation includes routers, Layer-2 and Layer-3 switches, Windows Server VMs, and the FortiGate firewall. It is used to validate connectivity, DNS, DHCP, authentication, NAT, and filtering rules before real deployment.

3.3 Non-Functional Requirements

Table 3: Non-Functional Requirement

Category	Description
Performance	- Fast user authentication with login response within seconds. - Stable DNS resolution with minimal latency. - Efficient IP address assignment through DHCP.
Security	- Strong password policies. - Firewall protection for internal and external traffic. - Strict access control based on least-privilege principles. [10]
Reliability	- Continuous operation of DNS and DHCP services without downtime. - Availability of backup mechanisms for critical data. - Support for Active Directory replication in future expansions.
Scalability	- Ability to easily accommodate new departments. - Support for additional VLANs and subnets. - Expansion of firewall rules.
Usability	- Clear and logical Organizational Unit (OU) structure. - Group Policy Objects (GPOs) easy to identify and modify. - Firewall policies well-labeled and organized for easy management.

3.4 Network Design Analysis [3]

1. **VLAN Segmentation**

To enhance security and reduce broadcast traffic, the network is logically divided into multiple VLANs. Each department operates within a dedicated VLAN to ensure isolation and controlled access.

2. **Routing and Communication Protocols**

Inter-VLAN communication is achieved through Layer-3 routing. In the proposed design, the **FortiGate Firewall** performs the role of the primary router for all VLANs to ensure that every packet passes through a security checkpoint before reaching other departments.

- **Inter-VLAN Routing via FortiGate to enforce security policies between departments.**



- **Optional dynamic routing protocols (such as RIP or OSPF) can be configured within GNS3 for simulation and testing future scalability.**

3. **Firewall Policy Analysis**

The FortiGate firewall plays a critical role in regulating traffic between VLANs and securing access to internal resources.

3.5 Software and Hardware Requirements

3.5.1 Software Requirements

- GNS3
- FortiGate
- VMware/VirtualBox
- Windows Server:
  - AD DS
  - DNS
  - DHCP
  - GPO

3.5.2 Hardware Requirements

- Virtual machines for Windows Server and network devices running inside GNS3.
- Stable internet connection for downloading tools, updates, and managing configurations.
- No physical hardware required except the administrator's PC or laptop.

3.6 Diagrams

1. Use Case Diagram

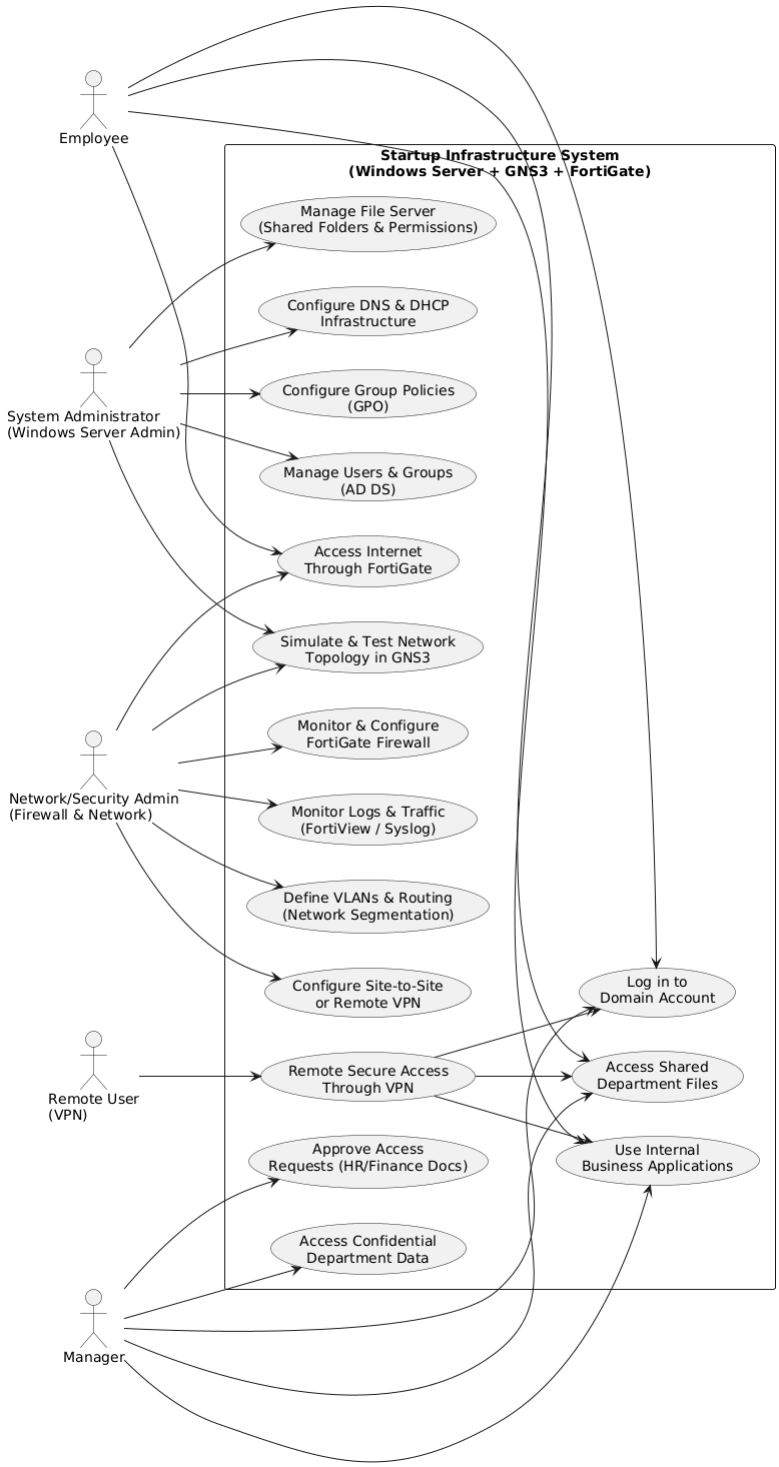


Figure 3: Use Case Diagram



- **Employee** users log in to the domain, access shared departmental files, and use internal business applications through centralized authentication.
- **The System Administrator manages** core server services, including Active Directory users and groups, DNS and DHCP configuration, Group Policy Objects (GPO), and file server permissions.
- **The Network/Security Administrator** is responsible for network security and connectivity. This includes configuring VLANs and routing, monitoring network traffic and logs, managing FortiGate firewall policies, and setting up VPN access.
- **Remote Users** securely access internal resources through VPN connections, ensuring protected remote connectivity.
- **Managers** have elevated privileges, allowing them to approve access requests and view confidential departmental data such as HR and Finance information.

---

2. Sequence Diagram

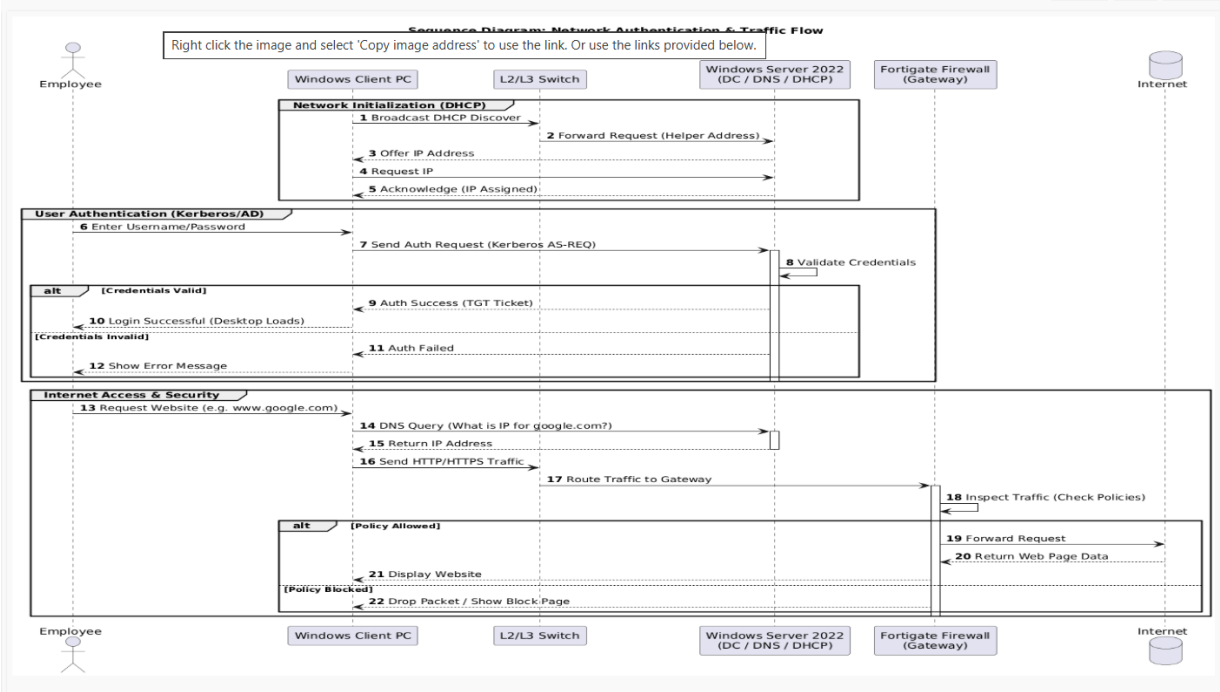


Figure 4: Sequence Diagram

1. Network Initialization (DHCP)
When the employee's device connects to the network:

1. The client broadcasts a **DHCP Discover** message.
2. The switch forwards the request to the DHCP server.
3. The server offers an available IP address.
4. The client requests the offered IP.
5. The server acknowledges and assigns the final IP configuration.

This process allows the device to join the internal network with valid IP, gateway, and DNS settings.

---

2. User Authentication (Kerberos / Active Directory)
After receiving network configuration:

6. The user enters their username and password.
7. The client sends an authentication request to Windows Server (Kerberos AS-REQ).
8. The Domain Controller validates the credentials.

Outcomes:

- **If credentials are valid:**
  9–10. The server issues a Kerberos ticket and the user logs in successfully.
- **If credentials are invalid:**
  11–12. Authentication fails and an error message is shown.

---

3. Internet Access & Security Filtering

Once logged in:

13. The user sends a web request (e.g., google.com).
    14–15. DNS resolves the website's IP address.
    16–17. The client sends HTTP/HTTPS traffic to the FortiGate firewall.

Firewall Processing:

18. FortiGate inspects the traffic and checks security policies.
19. If allowed, the firewall forwards the request to the Internet.
20. The web response returns to the user.

Outcomes:

- **If policy allows:**
  21. Website loads normally.
- **If policy blocks:**
  22. The firewall drops the packet and displays a block page.

---

3. Activity Diagram

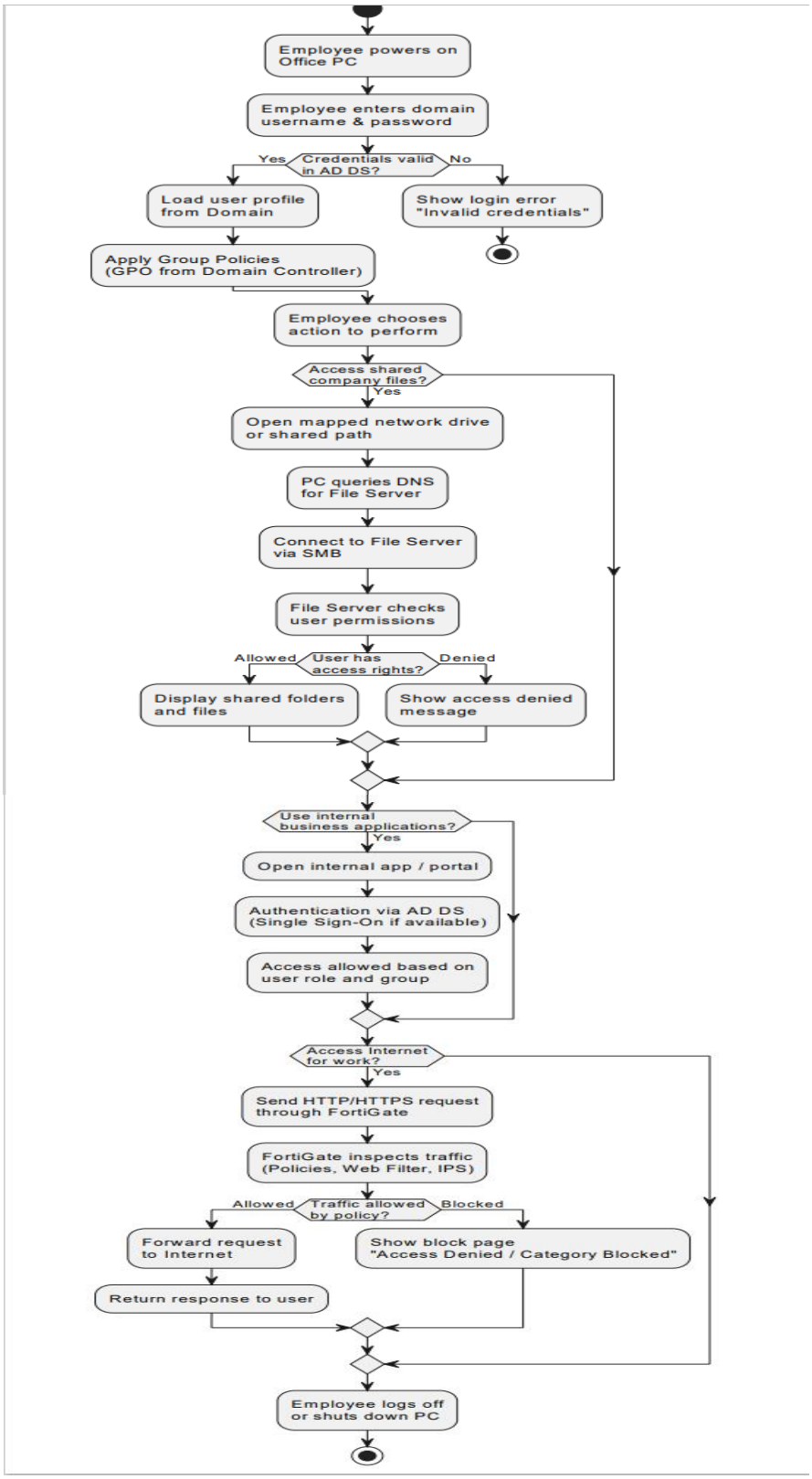


Figure 5: Activity Diagram

1. User Login Process

- The employee turns on their PC and enters domain credentials.
- The system verifies the credentials with Active Directory.
- If valid: The user profile and GPO settings are loaded.
- If invalid: A login error message is displayed.

2. Accessing Shared Files

- The employee chooses to open a mapped network drive or shared folder.
- The PC sends a DNS request and connects to the File Server.
- The File Server checks the user's permissions.
- If allowed: Shared folders and files are shown.
- If denied: An access denied message appears.

3. Internal Web Applications

- The employee may access an internal portal or web application.
- Authentication may occur again via AD DS (Single Sign-On if available).
- Access is granted or denied based on the user's role and group membership.

4. Internet Browsing

- The employee sends an HTTP/HTTPS request through the firewall.
- FortiGate inspects the traffic using policies, web filtering, and IPS.
- If allowed: The request is forwarded to the internet, and the response returns to the user.
- If blocked: A block page is displayed.

5. Logout

- The employee logs off or shuts down the computer, ending the workflow.

---

4. Data Flow Diagram

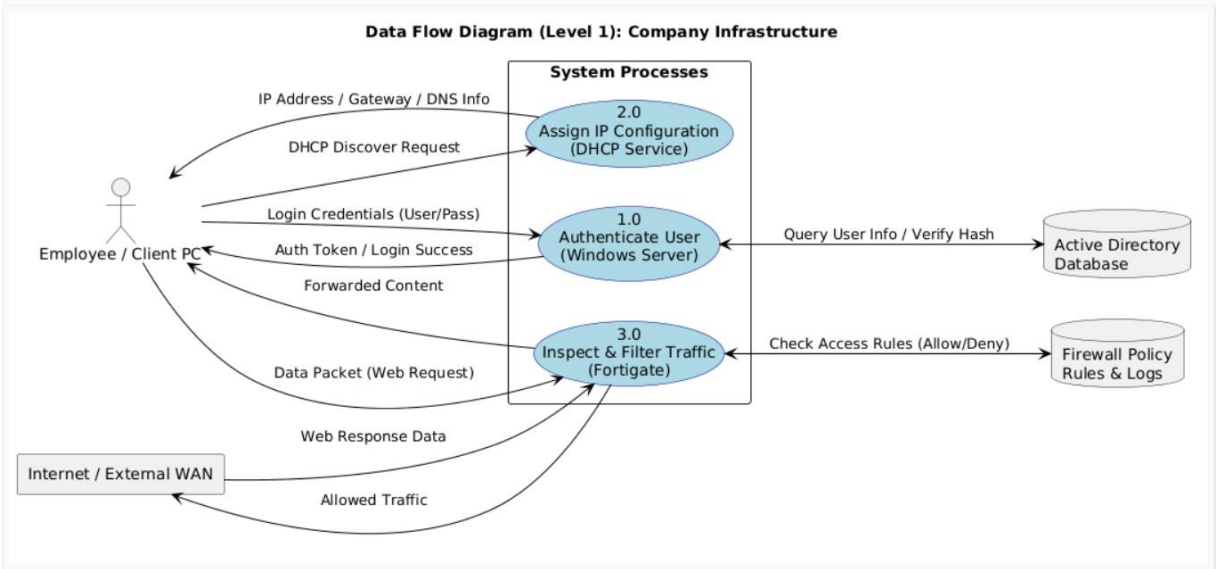


Figure 6: Data Flow Diagram

1. Assign IP Configuration

The employee connects to the network, the client PC sends a DHCP discovery request.
The DHCP server responds with:

- IP address
- Gateway
- DNS information

  This allows the device to communicate on the internal network.

2. Authenticate User

The employee sends login credentials (username/password) to the Windows Server.
The server checks the credentials by querying the Active Directory Database.
If successful, an authentication token is sent back to the user, allowing access to internal resources.

3. Inspect & Filter Traffic

When the employee sends a web request, the data packet passes through the FortiGate firewall.
The firewall:

- Checks security rules (allow/deny)
- Applies filtering and inspection
- Logs the traffic

Allowed traffic is forwarded to the external Internet, and the response is returned to the employee.

3.7 System Architecture Summary

- **Windows Server Domain Controller**
- **DNS/DHCP Services**
- **File Server**
- **FortiGate Firewall**
- **Layer-3 Switching (GNS3 simulation)**
- **Multiple VLANs**
- **Secure routing paths**

3.8 Code of Ethics

Ethical, Social, and Security considerations include protect employee privacy, preventing unauthorized monitoring, ensure role-based access, avoid security misconfigurations, complying with legal and corporate regulations.

3.9 Summary

This chapter presented a detailed analysis of the system requirements and network design. It covered functional and non-functional requirements, user roles, VLAN segmentation, routing, firewall policies, and security considerations. Diagrams were used to illustrate system behavior and data flow, forming the basis for the implementation phase.

Chapter 4

Design, Implementation and Testing

4.1 Overview

The main goal of this project is to **create, deploy, and test** a secure, scalable, and highly available enterprise network. The approach follows a **top-down network design methodology**, combining **Cisco routing and switching protocols**, **FortiGate security appliances**, and **Windows Server environments** to build a **centralized, resilient, and easy-to-manage IT infrastructure**

4.2 Design

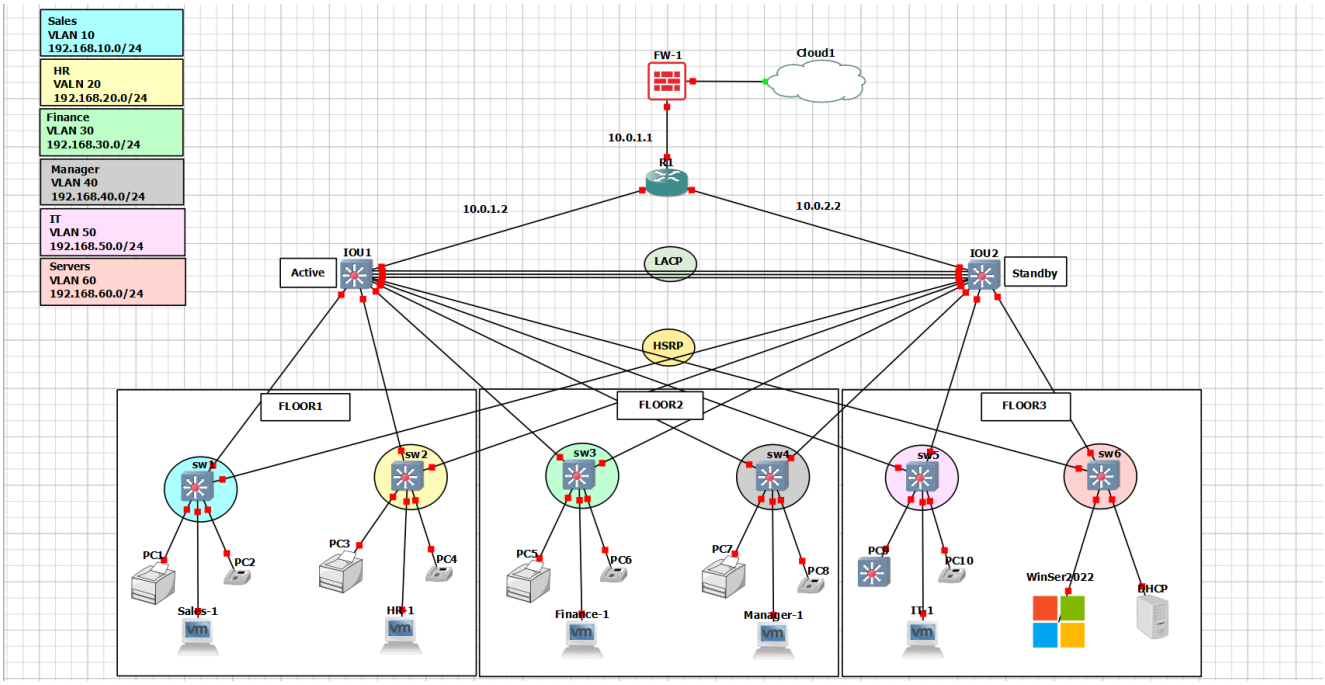


Figure 7: Full Network Topology

The network is designed using a **hierarchical architecture** to achieve both high performance and scalability:

- **Core and Distribution Layers:** Two multilayer switches, **IOU1** and **IOU2**, serve as the backbone of the network. They handle **inter-VLAN routing** and provide **redundancy**, ensuring continuous network operation even in case of a failure.
- **Access Layer:** Six access switches (Switches 1–6) connect end-user devices across multiple floors. The network is divided into **separate VLANs** (Sales, HR, Finance, Manager, IT, and Servers) to improve traffic management and maintain departmental data isolation.
- **Centralized Administration:** A **Windows Server 2022** manages **Active Directory (AD)**, **DNS**, and **DHCP**. Each VLAN has a dedicated DHCP scope, and all client PCs are successfully joined to the domain for centralized control and simplified management.
- **Real-Time Monitoring (NetWatch):** The **custom NetWatch dashboard** interfaces with the server to provide **live monitoring** of the network, including device status, ping latency, AD users, DHCP usage, and HSRP/firewall reachability (inferred from ICMP), enabling proactive oversight

4.3 Implementation

4.3.1 Used Protocols

- Hot Standby Router Protocol (HSRP):

HSRP was configured on the two core multilayer switches, **IOU1** and **IOU2**, to provide **gateway redundancy** and ensure high network availability.
Configuration: IOU1 was set as the **active gateway** with a higher priority, while IOU2 remained in **standby**, ready to take over automatically if IOU1 failed.
Benefit: This setup reduces network downtime and guarantees continuous connectivity for all VLANs across the enterprise.

- Rapid Spanning Tree Protocol (RSTP):

RSTP was deployed across the switching infrastructure to **prevent Layer 2 loops** and **broadcast storms**.
Configuration: IOU1 was designated as the **root bridge** by assigning it a lower spanning-tree priority, stabilizing traffic forwarding paths.
Benefit: RSTP improves network convergence time and enhances switching reliability during topology changes or link failures.

- Link Aggregation Control Protocol (LACP):

LACP was implemented between the core switches to **bundle multiple physical links into a single logical EtherChannel**.
Configuration: Multiple connections between IOU1 and IOU2 were aggregated to form one logical channel.
Benefit: This increases bandwidth, provides **link redundancy**, and ensures **fault tolerance**; if one physical link fails, the logical channel continues operating without interruption.

4.3.2 VLAN Segmentation

```
IOU1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et2/1, Et2/2, Et2/3, Et3/0
                                                Et3/1, Et3/2, Et3/3
10   Sales                            active
20   HR                               active
30   Finance                          active
40   Managere                         active
50   IT                               active
60   Servers                          active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
IOU1#
```

```
sw2#
sw2#
sw2#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Et0/0       on               802.1q         trunking      1
Et1/1       on               802.1q         trunking      1
Et1/2       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Et0/0       1,10,20,30,40,50,60
Et1/1       10,20,30,40,50,60
Et1/2       10,20,30,40,50,60

Port        Vlans allowed and active in management domain
Et0/0       1,20
Et1/1       20
Et1/2       20

Port        Vlans in spanning tree forwarding state and not pruned
Et0/0       1,20
Et1/1       20
Et1/2       20
```

Figure 8:VLAN Configuration

Virtual LANs (VLANs) were implemented to segment departmental traffic and improve security, scalability, and broadcast management within the enterprise network. VLANs were created to isolate departments and reduce broadcast traffic. Trunk links were configured between switches to allow VLAN communication across the infrastructure. [12]

Table 4:VLANs

VLAN ID	Department	Network	Gateway (HSRP)
10	Sales	192.168.10.0/24	192.168.10.1
20	HR	192.168.20.0/24	192.168.20.1
30	Finance	192.168.30.0/24	192.168.30.1
40	Manager	192.168.40.0/24	192.168.40.1
50	IT	192.168.50.0/24	192.168.50.1
60	Servers	192.168.60.0/24	192.168.60.1

4.3.3 IOU1

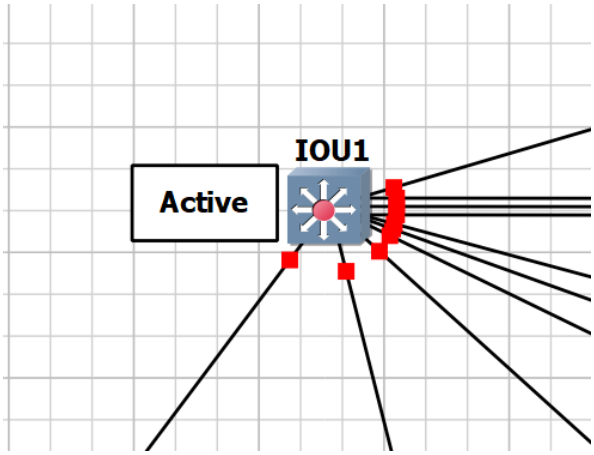


Figure 9:IOU1

1. STP

```
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
spanning-tree vlan 1,10,20,30,40,50,60 priority 4096
!
```

IOU1 IOU2

```
IOU1#show spanning-tree vlan 20

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID     Priority    4116
              Address     aabb.cc00.0200
              This bridge is the root
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    4116   (priority 4096 sys-id-ext 20)
              Address     aabb.cc00.0200
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/1               Desg FWD 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr
Et1/0               Desg FWD 100       128.5    Shr
Et1/1               Desg FWD 100       128.6    Shr
Et1/2               Desg FWD 100       128.7    Shr
Po1                 Desg FWD 56        128.65   Shr


IOU1#
```

Figure 10:Spanning-tree configuration

The switch priority was manually lowered to 4096 to ensure that IOU1 operates as the root bridge for all VLANs, providing stable Layer 2 forwarding and preventing switching loops within the enterprise network.

---

2. LACP EtherChannel Configuration

```
!
interface Port-channel1
 switchport trunk allowed vlan 10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

```
interface Ethernet1/3
 switchport trunk allowed vlan 10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface Ethernet2/0
 switchport trunk allowed vlan 10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
```

Figure 11:EtherChannel configuration

Multiple physical links were bundled into a single logical Port-Channel interface to provide redundancy, load balancing, and higher availability between the distribution switches.

---

3. VLAN Trunking

```
!
interface Ethernet0/1
 description TO-SWITCH1
 switchport access vlan 10
 switchport trunk allowed vlan 1,10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/2
 description TO-SWITCH2
 switchport access vlan 20
 switchport trunk allowed vlan 1,10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
```

Figure 12:Trunking

the trunk port configuration used to connect the core switch with the access switches.
IEEE 802.1Q trunking was configured to allow VLAN traffic to traverse between network layers while maintaining VLAN segmentation across the infrastructure.

---

4. Inter-VLAN Routing + HSRP

```
interface Vlan10
 description Sales-SVI
 ip address 192.168.10.2 255.255.255.0
 ip helper-address 192.168.60.20
 standby 10 ip 192.168.10.1
 standby 10 priority 120
 standby 10 preempt
!
interface Vlan20
 description HR-SVI
 ip address 192.168.20.2 255.255.255.0
 ip helper-address 192.168.60.20
 standby 20 ip 192.168.20.1
 standby 20 priority 120
 standby 20 preempt
!
interface Vlan30
 description Finance-SVI
 ip address 192.168.30.2 255.255.255.0
 ip helper-address 192.168.60.20
 standby 30 ip 192.168.30.1
 standby 30 priority 120
 standby 30 preempt
```

Figure 13:SVI

demonstrates the implementation of inter-VLAN routing using Switch Virtual Interfaces (SVIs).
HSRP was configured to provide gateway redundancy for each VLAN, ensuring continuous network connectivity in the event of a core switch failure.
DHCP relay functionality was also configured using the IP helper-address command to forward DHCP requests to the centralized DHCP server.

---

5. Static Route

```
no ip http secure-server
!
ip route 0.0.0.0 0.0.0.0 10.0.1.1
ip route 0.0.0.0 0.0.0.0 192.168.60.20
!
!
```

Figure 14: static route

shows the default static route configured on the core switch to forward external traffic toward the perimeter firewall and upstream routing infrastructure.

4.3.4 IOU2

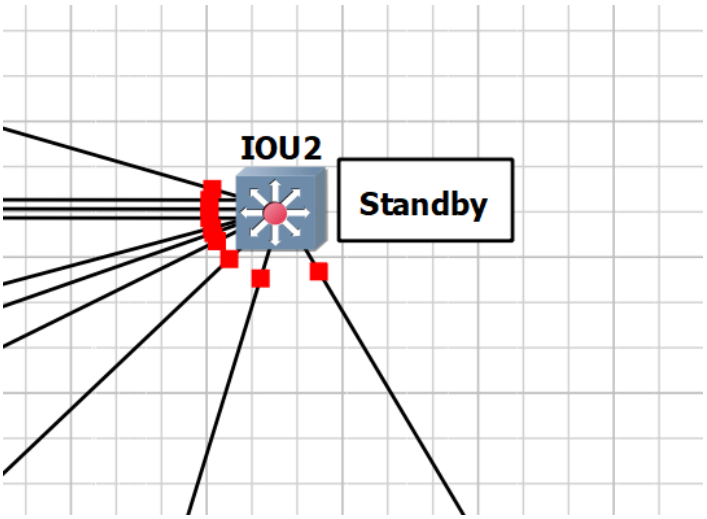


Figure 15:IOU2

1. **Spanning Tree Backup Root**

```
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
spanning-tree vlan 1,10,20,30,40,50,60 priority 8192
!
```

Figure 16:STP Configuration

Rapid PVST was implemented to prevent switching loops and improve convergence time.
IOU2 was configured with a secondary spanning-tree priority value to act as the backup root bridge if IOU1 becomes unavailable.

---

2. **Core Switch Redundancy Configuration (HSRP)**

```
interface Vlan10
 description Sales-SVI
 ip address 192.168.10.3 255.255.255.0
 ip helper-address 192.168.60.20
 standby 10 ip 192.168.10.1
 standby 10 preempt
!
interface Vlan20
 description HR-SVI
 ip address 192.168.20.3 255.255.255.0
 ip helper-address 192.168.60.20
 standby 20 ip 192.168.20.1
 standby 20 preempt
!
```

Figure 17:Gateway Redundancy using HSRP

HSRP was configured on IOU2 to provide gateway redundancy for all VLANs.
The virtual gateway IP addresses were shared between IOU1 and IOU2 to ensure continuous network availability in case the primary core switch fails.
The “preempt” command allows the preferred switch to regain the active role automatically after recovery. [14]

3. LACP EtherChannel

```
!
interface Port-channel1
 switchport trunk allowed vlan 10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet1/3
 switchport trunk allowed vlan 10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface Ethernet2/0
 switchport trunk allowed vlan 10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
```

Figure 18:Link Aggregation using LACP

LACP EtherChannel was configured between IOU1 and IOU2 to provide link redundancy and increased bandwidth.
Multiple physical links were combined into a single logical Port-Channel interface to improve fault tolerance and traffic distribution.

4. VLAN Trunking

```
!
interface Ethernet2/1
 switchport trunk allowed vlan 10,20,30,40,50,60
 switchport trunk encapsulation dot1q
 switchport mode trunk
 shutdown
!
```

Figure 19:VLAN Trunk Configuration

802.1Q trunk links were configured between the core and access switches to allow VLAN traffic to pass across the network infrastructure.
Only the required VLANs were permitted on the trunk interfaces to improve network organization and security.

5. DHCP Relay

```
!
interface Vlan20
 description HR-SVI
 ip address 192.168.20.3 255.255.255.0
 ip helper-address 192.168.60.20
 standby 20 ip 192.168.20.1
 standby 20 preempt
!
```

Figure 20:DHCP Configuration

DHCP relay functionality was configured using the IP helper-address command to forward DHCP broadcast requests from multiple VLANs to the centralized DHCP server.

6. Static Default Route

```
!
ip route 0.0.0.0 0.0.0.0 10.0.2.1
ip route 0.0.0.0 0.0.0.0 192.168.60.20
!
```

Figure 21:Default Routing Configuration

A default static route was configured on IOU2 to forward unknown traffic toward the upstream router and firewall infrastructure for external connectivity.

4.3.5 R1

1. Transit Network Interfaces

```
!
interface FastEthernet0/0
 ip address 10.0.0.2 255.255.255.252
 duplex full
!
interface FastEthernet1/0
 ip address 10.0.1.1 255.255.255.252
 duplex half
!
interface FastEthernet2/0
 ip address 10.0.2.1 255.255.255.252
 duplex auto
 speed auto
!
```

Figure 22:Transit Routing Infrastructure

R1 was configured as a transit router between the FortiGate firewall and the multilayer core switches.
Dedicated point-to-point transit networks were implemented to establish routed communication paths with IOU1 and IOU2.

---

2. Default Route Toward Firewall

```
 negotiation auto
!
ip forward-protocol nd
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

Figure 23:Upstream Default Routing

A default static route was configured on R1 to forward external traffic toward the FortiGate firewall for internet access and perimeter security processing.

---

3. Floating Static Routes

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1
ip route 192.168.10.0 255.255.255.0 10.0.1.2
ip route 192.168.10.0 255.255.255.0 10.0.2.2 10
ip route 192.168.20.0 255.255.255.0 10.0.1.2
ip route 192.168.20.0 255.255.255.0 10.0.2.2 10
ip route 192.168.30.0 255.255.255.0 10.0.1.2
ip route 192.168.30.0 255.255.255.0 10.0.2.2 10
ip route 192.168.40.0 255.255.255.0 10.0.1.2
ip route 192.168.40.0 255.255.255.0 10.0.2.2 10
ip route 192.168.50.0 255.255.255.0 10.0.1.2
ip route 192.168.50.0 255.255.255.0 10.0.2.2 10
ip route 192.168.60.0 255.255.255.0 10.0.1.2
ip route 192.168.60.0 255.255.255.0 10.0.2.2 10
```

Figure 24:Floating Static Route Failover

Floating static routes were implemented on R1 to provide routing redundancy between the primary and backup multilayer switches.
Backup routing paths were configured for all VLAN networks to ensure continuous communication during link or device failures.

4.3.6 FortiGate Firewall

1. WAN / LAN Interface Configuration

```
CLI Console (1)
FortiGate-VM64 # show system interface
config system interface
    edit "port1"
        set vdom "root"
        set mode dhcp
        set allowaccess ping https ssh http
        set type physical
        set lldp-reception enable
        set role wan
        set snmp-index 1
    next
    edit "port2"
        set vdom "root"
        set ip 10.0.0.1 255.255.255.252
        set allowaccess ping https ssh http
        set type physical
        set device-identification enable
        set lldp-transmission enable
        set role lan
        set snmp-index 2
    next
```

Figure 25:FortiGate Interface Configuration

FortiGate interfaces were configured to separate external and internal network segments.
Port1 was assigned as the WAN interface connected to the internet cloud, while Port2 was configured as the internal transit interface connected to the routing infrastructure.

---

2. Static Routes Toward VLANs

CLI Console (1)

```
FortiGate-VM64 #
FortiGate-VM64 # show router static
config router static
    edit 2
        set dst 192.168.10.0 255.255.255.0
        set gateway 10.0.0.2
        set device "port2"
    next
    edit 3
        set dst 192.168.20.0 255.255.255.0
        set gateway 10.0.0.2
        set device "port2"
    next
    edit 4
```

Figure 26:Internal VLAN Routing

```
FortiGate-VM64 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default

Routing table for VRF=0
S*      0.0.0.0/0 [5/0] via 192.168.1.1, port1, [1/0]
C       10.0.0.0/30 is directly connected, port2
C       192.168.1.0/24 is directly connected, port1
S       192.168.10.0/24 [10/0] via 10.0.0.2, port2, [1/0]
S       192.168.20.0/24 [10/0] via 10.0.0.2, port2, [1/0]
S       192.168.30.0/24 [10/0] via 10.0.0.2, port2, [1/0]
S       192.168.40.0/24 [10/0] via 10.0.0.2, port2, [1/0]
S       192.168.50.0/24 [10/0] via 10.0.0.2, port2, [1/0]
S       192.168.60.0/24 [10/0] via 10.0.0.2, port2, [1/0]


FortiGate-VM64 #
```

Figure 27:Routing Table

Static routes were configured on the FortiGate firewall to enable communication with internal VLAN networks through the transit router.

3. IT & Manager Policies

CLI Console (1)

```
FortiGate-VM64 # show firewall policy
config firewall policy
    edit 1
        set name "IT-Full-Access"
        set uuid 6c366aa2-4694-51f1-5123-bb9db5a6cbaa
        set srcintf "port2"
        set dstintf "port1"
        set action accept
        set srcaddr "IT-Net"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set ssl-ssh-profile "certificate-inspection"
        set nat enable
    next
    edit 2
        set name "Manager-Full-Access"
        set uuid 95a5edae-4694-51f1-8b51-caedf4421e68
        set srcintf "port2"
        set dstintf "port1"
        set action accept
        set srcaddr "Manager-Net"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set ssl-ssh-profile "certificate-inspection"
        set nat enable
```

Figure 28:IT & Manager Policy

4. HR & Sales Policies

CLI Console (1)

```
        set nat enable
    next
    edit 3
        set name "HR-Internet"
        set uuid 0928a956-4695-51f1-2dd5-0ab6d5ed9ec8
        set srcintf "port2"
        set dstintf "port1"
        set action accept
        set srcaddr "HR-Net"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set ssl-ssh-profile "certificate-inspection"
        set nat enable
    next
    edit 4
        set name "Sales-Internet"
        set uuid 6ca5994e-4695-51f1-3a8b-fcaf59955911
        set srcintf "port2"
        set dstintf "port1"
        set action accept
        set srcaddr "Sales-Net"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set ssl-ssh-profile "certificate-inspection"
        set nat enable
```

Figure 29:HR & Sales Policy

5. Finance & Block Policies

CLI Console (1)

```
    next
    edit 5
        set name "Finance-Servers"
        set uuid 47738270-4696-51f1-2717-42b371510c12
        set srcintf "port2"
        set dstintf "port2"
        set action accept
        set srcaddr "Finance-Net"
        set dstaddr "Servers-Net"
        set schedule "always"
        set service "ALL"
        set ssl-ssh-profile "certificate-inspection"
    next
    edit 6
        set name "Block-Inter-VLAN"
        set uuid c9f4fc2e-4696-51f1-3edf-67e0b6b88804
        set srcintf "port2"
        set dstintf "port2"
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set logtraffic disable
    next
```

Figure 30:Finance & Block Policy

6. LAN-TO-WAN Policy

CLI Console (1)

```
        set service "ALL"
        set logtraffic disable
    next
    edit 7
        set name "LAN-to-WAN"
        set uuid e935c296-48ba-51f1-dadb-8ece304a16f7
        set srcintf "port2"
        set dstintf "port1"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set nat enable
    next
    edit 8
        set name "LAN-to-WAN-FIX"
        set uuid bf59e732-48d2-51f1-2a6a-373904e3a1b0
        set srcintf "port2"
        set dstintf "port1"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set nat enable
    next
```

Figure 31:Firewall Security Policies

Firewall policies were configured to control traffic flow between internal VLANs and the external network.
Network Address Translation (NAT) was enabled to allow internal devices to access internet services securely.

4.3.7 DHCP and IP Address Management

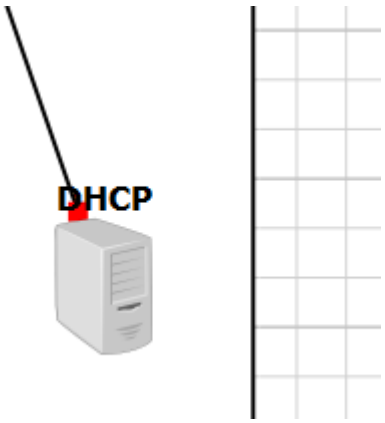


Figure 32:DHCP

```
DHCP#
DHCP#
DHCP#show running-config | include dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.9
ip dhcp excluded-address 192.168.10.201 192.168.10.254
ip dhcp excluded-address 192.168.20.1 192.168.20.9
ip dhcp excluded-address 192.168.20.201 192.168.20.254
ip dhcp excluded-address 192.168.30.1 192.168.30.9
ip dhcp excluded-address 192.168.30.201 192.168.30.254
ip dhcp excluded-address 192.168.40.1 192.168.40.9
ip dhcp excluded-address 192.168.40.201 192.168.40.254
ip dhcp excluded-address 192.168.50.1 192.168.50.9
ip dhcp excluded-address 192.168.50.201 192.168.50.254
ip dhcp excluded-address 192.168.60.1 192.168.60.9
ip dhcp excluded-address 192.168.60.201 192.168.60.254
ip dhcp excluded-address 192.168.60.20
ip dhcp excluded-address 192.168.60.100
ip dhcp pool VLAN10
ip dhcp pool VLAN20
ip dhcp pool VLAN30
ip dhcp pool VLAN40
ip dhcp pool VLAN50
ip dhcp pool VLAN60
DHCP#
```

Figure 33:DHCP Configuration

Dynamic Host Configuration Protocol (DHCP) services were configured on the router to automatically assign IP addresses, default gateways, and DNS settings for all VLANs in the enterprise network.

4.3.8 Windows Server and Active Directory

1. **Windows Server 2022**
   machine was deployed to provide centralized network services including Active Directory Domain Services (AD DS), DNS, and domain-based authentication.

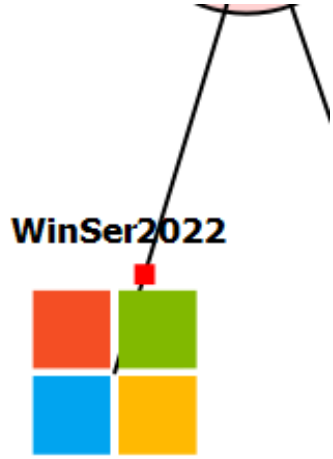


Figure 34:WinServer2022

2. **Active Directory**

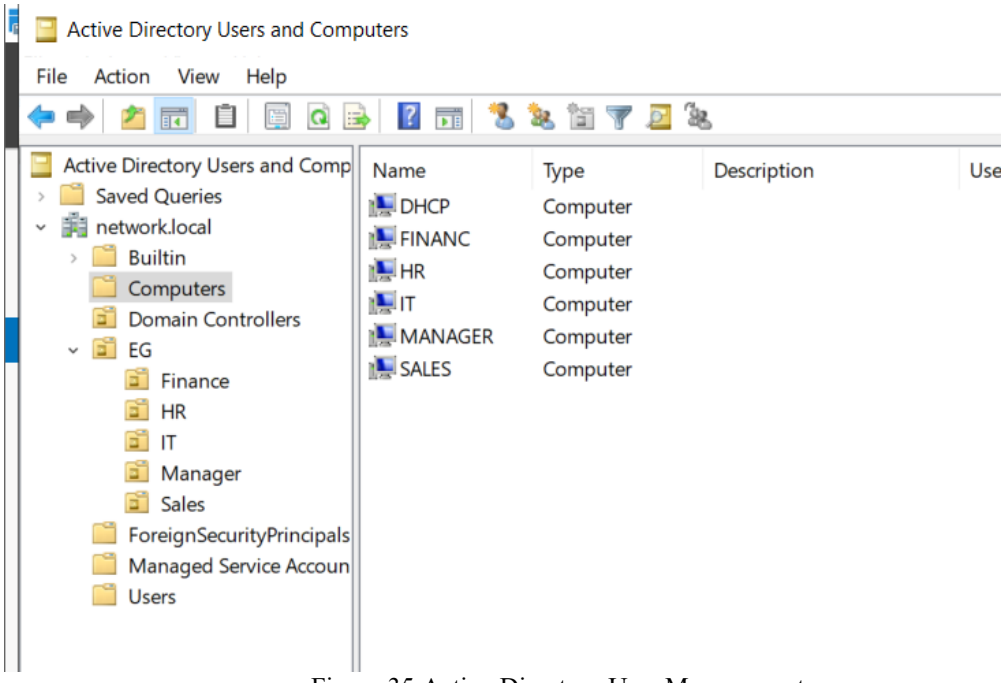


Figure 35:Active Directory User Management

Active Directory Domain Services (AD DS) was deployed to provide centralized authentication and user management across the enterprise network infrastructure.

3. DNS

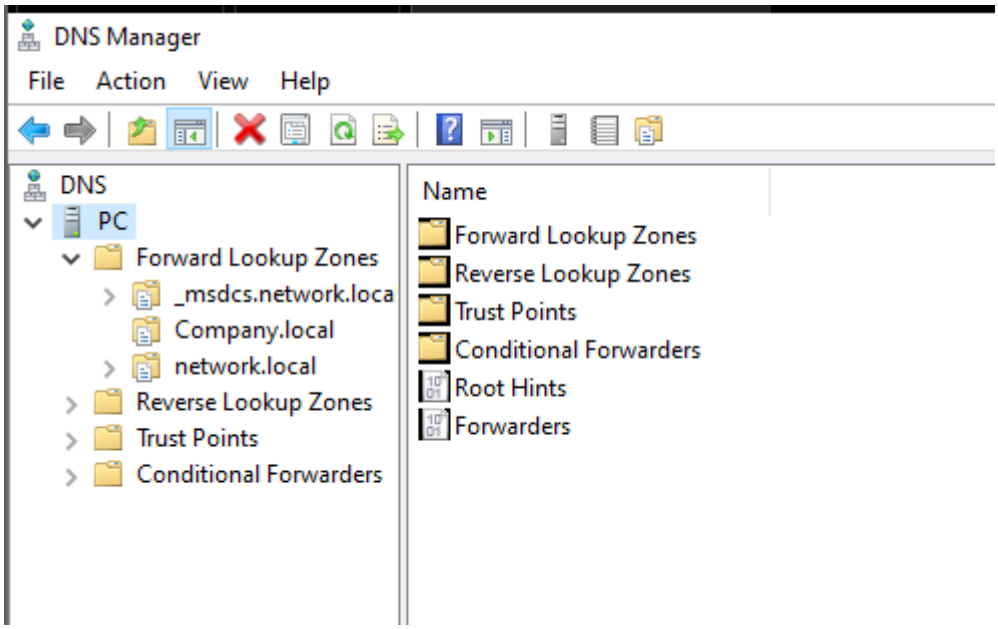


Figure 36:DNS Service Configuration

The DNS server was configured to provide internal name resolution services for domain-joined devices and network services.

---

4. GPO

Group Policy Objects (GPOs) were implemented to enforce security policies across all domain-joined devices

- Password policy
- Desktop restrictions
- Disable USB
- Firewall rules

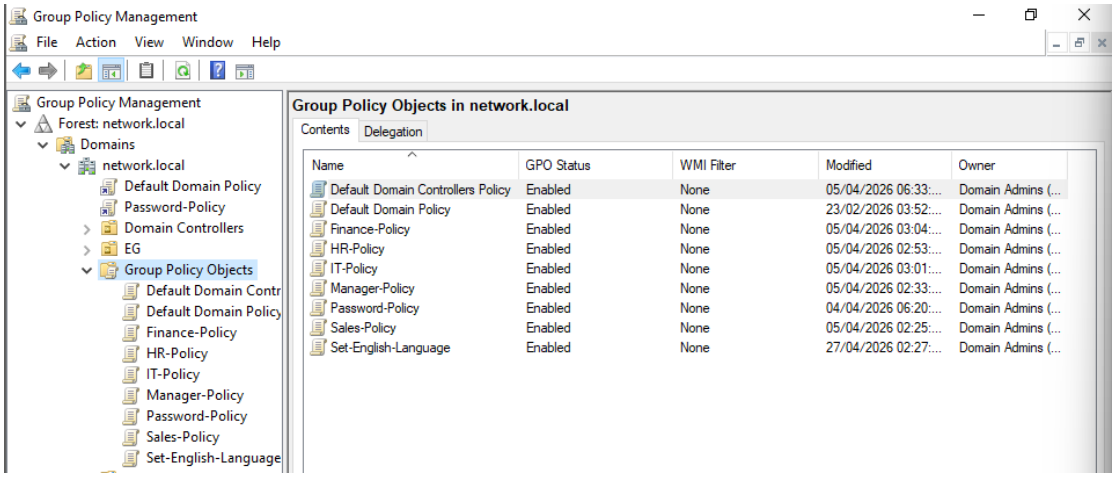


Figure 37:GPO Configuration

Group Policy Objects (GPOs) were implemented to centrally manage domain security configurations and enforce password policies across all authenticated users and systems.

5. Password settings

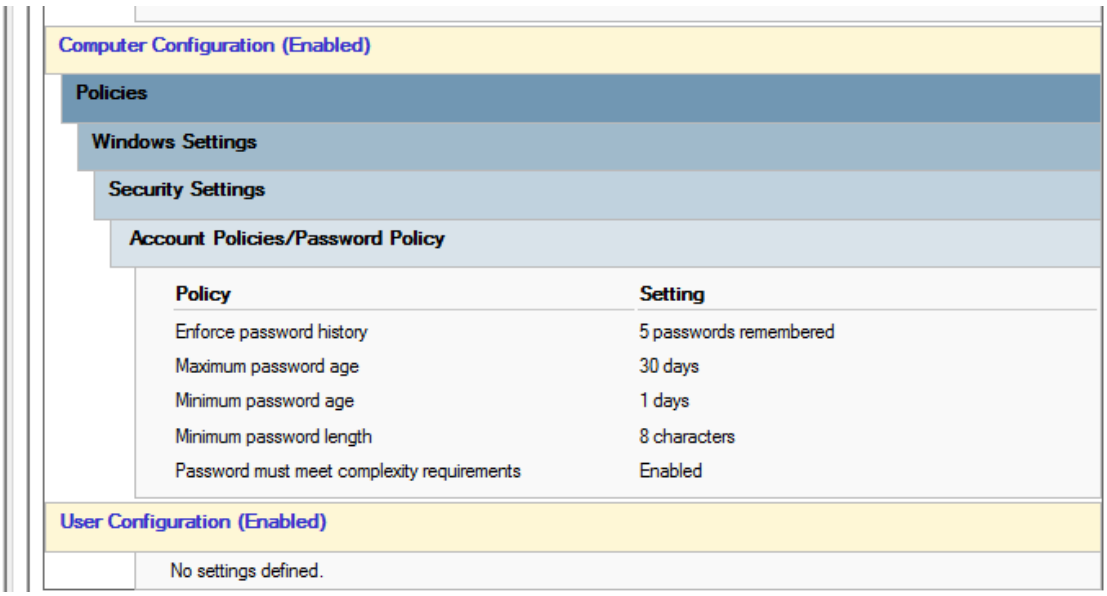


Figure 38:Password Security Policy

A centralized password policy was configured using Group Policy Management to enforce secure authentication requirements within the enterprise domain environment.

Table 5:Services

Service	Location	Function
AD DS	Windows Server	Authentication
DNS	Windows Server	Name Resolution
DHCP	Router	IP Assignment
GPO	Windows Server	Security Policies

4.4 Dashboard Design and Implementation

A custom real-time monitoring dashboard **named NetWatch** was developed to provide centralized visibility of the network infrastructure.
The dashboard monitors network devices (ICMP), Active Directory users, DHCP pool usage, and firewall/core-switch reachability in near real time.
The main objective of the dashboard is to simplify fault detection and provide live infrastructure monitoring through a modern web-based interface.

4.4.1 Dashboard Frontend Design

The dashboard frontend was developed using HTML, CSS, and JavaScript.
A dark cyber-security inspired interface was designed to provide a professional SIEM-like monitoring experience. [11]
The interface includes:

- Device Monitoring Panel
- Topology Visualization
- Fault Detection Engine
- Active Directory User Monitoring
- DHCP Monitoring
- Simulated Traffic Chart (demo)
- Auto Refresh Mechanism

Login Page:

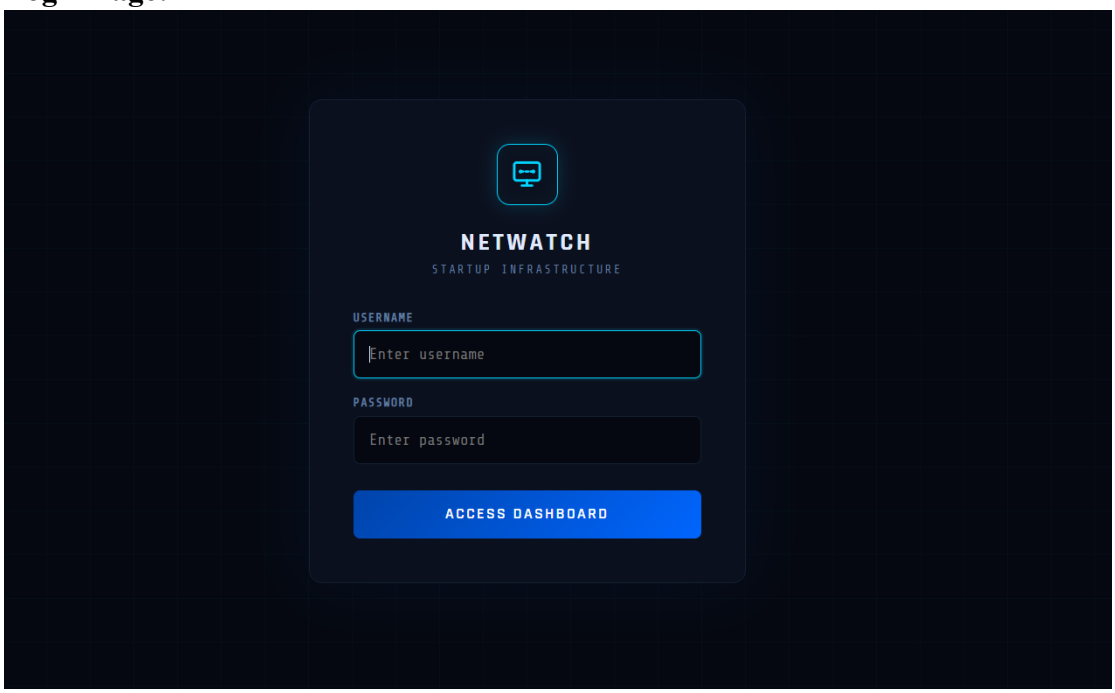


Figure 39:NetWatch Login Interface

The NetWatch monitoring system provides a secure authentication interface for administrators before accessing the monitoring dashboard. The interface was developed using HTML, CSS, and JavaScript. [15]

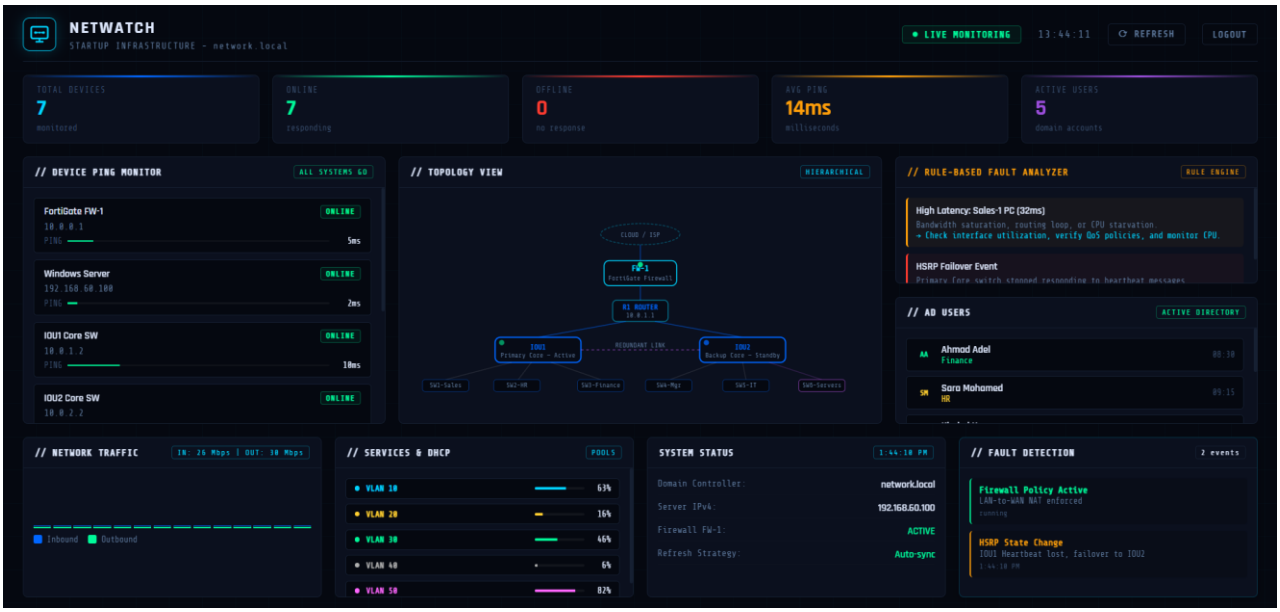


Figure 40:NetWatch Real-Time Monitoring Dashboard

4.4.2 Technologies Used

Table 6:Technologies used

Technology	Purpose
HTML	Dashboard Structure
CSS	User Interface Styling
JavaScript	Dynamic Data Rendering
Python	Backend Monitoring Engine
IIS	Web Hosting
PowerShell	Active Directory Integration
JSON	Real-Time Data Exchange

4.4.3index.html

1. Metrics Cards

```
78      <div class="dashboard-layout">
79
80          <div class="metrics">
81              <div class="metric-card glass-panel blue">
82                  <div class="metric-label">Total Devices</div>
83                  <div class="metric-value blue" id="m-total">--</div>
84                  <div class="metric-sub">monitored</div>
85              </div>
86              <div class="metric-card glass-panel green">
87                  <div class="metric-label">Online</div>
88                  <div class="metric-value green" id="m-online">--</div>
89                  <div class="metric-sub">responding</div>
90              </div>
91              <div class="metric-card glass-panel red">
92                  <div class="metric-label">Offline</div>
93                  <div class="metric-value red" id="m-offline">--</div>
94                  <div class="metric-sub">no response</div>
95              </div>
96              <div class="metric-card glass-panel amber">
97                  <div class="metric-label">Avg Ping</div>
98                  <div class="metric-value amber" id="m-ping">--</div>
99                  <div class="metric-sub">milliseconds</div>
100             </div>
101             <div class="metric-card glass-panel purple">
102                 <div class="metric-label">Active Users</div>
103                 <div class="metric-value purple" id="m-users">--</div>
104                 <div class="metric-sub">domain accounts</div>
105             </div>
106         </div>
```

Figure 41:Dynamic metrics

Dynamic metric cards were implemented to display real-time infrastructure statistics including online devices, offline systems, average latency, and active domain users.

2. Device Monitoring Section

```
108
109         <div class="middle-section">
110
111             <div class="glass-panel panel-flex">
112                 <div class="panel-header">
113                     <div class="panel-title">// Device Ping Monitor</div>
114                     <span class="tag green" id="ping-status">LOADING</span>
115                 </div>
116                 <div class="panel-body scrollable-y">
117                     <div class="devices-grid" id="devices-grid">
118                         <div class="loading">Scanning devices...</div>
119                     </div>
120                 </div>
121             </div>
```

Figure 42:Device Monitoring

A dedicated device monitoring panel was implemented to display device availability and ICMP latency values dynamically.

3. Fault Detection Panel

```
300
301                <div class="glass-panel panel-flex">
302                    <div class="panel-header">
303                        <div class="panel-title">// Fault Detection</div>
304                        <span class="tag" id="alert-count">-- events</span>
305                    </div>
306                    <div class="panel-body scrollable-y">
307                        <div class="alerts-list" id="alerts-list">
308                            <div class="loading">Loading alerts...</div>
309                        </div>
```

Figure 43:Fault Detection

The dashboard includes a fault detection module responsible for displaying generated alerts related to offline devices and infrastructure failures.

4.4.4 app.js

1. VLAN Definition

```
1  const vlans = [
2    { name: 'Sales', id: 10, subnet: '192.168.10.0/24', gw: '192.168.10.254', color: 'var(--sales)' },
3    { name: 'HR', id: 20, subnet: '192.168.20.0/24', gw: '192.168.20.254', color: 'var(--hr)' },
4    { name: 'Finance', id: 30, subnet: '192.168.30.0/24', gw: '192.168.30.254', color: 'var(--finance)' },
5    { name: 'Manager', id: 40, subnet: '192.168.40.0/24', gw: '192.168.40.254', color: 'var(--manager)' },
6    { name: 'IT', id: 50, subnet: '192.168.50.0/24', gw: '192.168.50.254', color: 'var(--it)' },
7    { name: 'Servers', id: 60, subnet: '192.168.60.0/24', gw: '192.168.60.254', color: 'var(--servers)' },
8  ];
```

Figure 44:VLAN Definition

VLAN information was defined programmatically to represent the segmented enterprise network architecture.

2. Dynamic JSON Loading

```
20   async function loadData() {
21     try {
22       const res = await fetch('data.json?t=' + Date.now());
23       if (!res.ok) throw new Error('data.json not found');
24       const data = await res.json();
25       isDemoMode = false;
26       renderData(data);
```

Figure 45:Dynamic JSON

JavaScript asynchronous requests were used to retrieve live monitoring data from the generated JSON backend without reloading the webpage.

3. Rule-Based Fault Analyzer

```
111
112  // RULE-BASED FAULT ANALYZER
113  function renderRuleBasedFaultAnalyzer(devices, alerts) {
114    const aiList = document.getElementById('ai-fault-list');
115    let analysis = [];
116
117    // Rule 1: Offline Devices
118    const offline = devices.filter(d => d.status === 'offline');
119    offline.forEach(d => {
120      analysis.push({
121        type: 'critical',
122        issue: `Device Offline: ${d.name}`,
123        cause: 'VLAN misconfiguration, power loss, or physical link failure.',
124        action: 'Check switch port status, verify IP configuration, and test physical connectivity.'
125      });
126    });
```

Figure 46:Fault Analyzer

A rule-based fault analyzer was implemented to detect infrastructure anomalies such as offline devices and abnormal latency conditions. [16]

4. Auto Refresh

```
349
350    window.dashIntervals = [
351      setInterval(loadData, 10000), // Speed up mock for dynamic testing
352      setInterval(renderTraffic, 4000),
353      setInterval(updateClock, 1000)
354    ];
355  }
```

Figure 47:Auto Refresh

Automatic refresh intervals were implemented to provide continuous live monitoring updates.

5. Login Authentication

```
300  const VALID_USERS = [
301    { username: 'admin', password: '1234@' },
302    { username: 'roaa', password: '1234@' },
303    { username: 'network', password: '1234@' },
304  ];
305
```

Figure 48:Login Authentication

A lightweight authentication mechanism was implemented to restrict unauthorized dashboard access.

4.4.5 network_monitor.py

A Python-based backend monitoring engine was developed to collect live network information and generate dynamic monitoring data.
The monitoring engine continuously scans network devices using ICMP ping requests and updates a JSON data file consumed by the dashboard frontend.

1. **Device Definitions**

```
16  DEVICES = [
17      {"name": "FortiGate FW-1",   "ip": "10.0.0.1"},
18      {"name": "Windows Server",   "ip": "192.168.60.100"},
19      {"name": "IOU1 Core SW",     "ip": "10.0.1.2"},
20      {"name": "IOU2 Core SW",     "ip": "10.0.2.2"},
21      {"name": "R1 Router",        "ip": "192.168.60.20"},
22      {"name": "Sales-1 PC",       "ip": "192.168.10.10"},
23      {"name": "HR-1 PC",       "ip": "192.168.20.10"},
24      {"name": "Finance-1 PC",     "ip": "192.168.30.10"},
25      {"name": "Manager-1 PC",       "ip": "192.168.40.10"},
26      {"name": "IT-1 PC",       "ip": "192.168.50.10"}
27  ]
```

Figure 49:Device Definitions

The Python backend maintains a list of monitored infrastructure devices including firewalls, servers, switches, and end-user systems.

2. **ICMP Ping Monitoring**

```
40  def ping_device(device, output_list):
41      """Pings a single device and records the latency/status."""
42      ip = device['ip']
43      cmd = ["ping", "-n", "1", "-w", "1000", ip]
44      try:
45          startupinfo = subprocess.STARTUPINFO()
46          startupinfo.dwFlags |= subprocess.STARTF_USESHOWWINDOW
47
48          output = subprocess.check_output(cmd, stderr=subprocess.STDOUT, universal_newlines=True, startupinfo=startupinfo)
49
50          if "TTL=" in output:
51              status = "online"
52              try:
53                  time_str = [line for line in output.split('\n') if "time=" in line or "time<" in line][0]
54                  if "time<" in time_str:
55                      ping_ms = 1
```

Figure 50:ICMP Ping Monitoring

The Python subprocess module was used to execute ICMP ping commands against monitored devices.
The system sends a single ping request to each device and analyzes the response to determine device availability and network latency.

Table 7:Ping

Parameter	Function
-n 1	Sends one ICMP packet
-w 1000	**Waits 1 second before timeout**

3. Alert Generation

```
156
157     for dev in device_results:
158         if dev["status"] == "offline":
159             alerts.append({
160                 "type": "err",
161                 "title": f"Device Offline: {dev['name']}",
162                 "msg": f"IP {dev['ip']} is not responding to ICMP.",
163                 "time": datetime.now().strftime("%I:%M:%S %p")
164             })
```

Figure 51:Alert Generation

The monitoring backend automatically generated alerts whenever monitored devices became unreachable.

4. JSON Generation

```
213     }
214     temp_file = OUTPUT_FILE + ".tmp"
215     with open(temp_file, "w") as f:
216         json.dump(results, f, indent=4)
217     os.replace(temp_file, OUTPUT_FILE)
```

Figure 52:JSON Generation

Monitoring results were continuously exported into a structured JSON file used by the frontend dashboard.

4.4.6 Real-Time JSON Data Generation

The backend monitoring engine dynamically generates a JSON file every few seconds.
The generated JSON file contains:

- **Device status**
- **Ping response time**
- **DHCP statistics**
- **Active Directory users**
- **Fault alerts**
- **Data-source flags (live / sample)**

4.4.7 Dashboard Features Summary

Table 8:Dashboard summary

Feature	Description
Live Device Monitoring	Detects online/offline devices
Fault Detection	Generates alerts
AD Monitoring	Displays domain users
DHCP Monitoring	Shows VLAN usage
Topology View	Visual network hierarchy
Traffic Chart (demo)	Simulated traffic graph
Auto Refresh	Real-time updates

The implemented dashboard successfully provided centralized infrastructure monitoring and simplified fault detection across the enterprise network environment. The integration between Python, PowerShell, IIS, and JavaScript enabled near real-time monitoring capabilities through a modern web interface.

4.5 Tools and Libraries

Name	Icon	Description
GNS3		It is a free, open-source software that emulates, configures, and tests virtual and real networks.
VMware Workstation		It is a leading provider of virtualization software that enables organizations to run multiple operating systems and applications on a single physical server.
Visual studio code		It is a free, lightweight editor by Microsoft that supports debugging, syntax highlighting, and language extensions.
Python		Python is a high-level, interpreted programming language known for its simple syntax, readability, and versatility. It is widely used for web development, data science, AI, and automation.
Microsoft Word		Microsoft word is Word processing used for preparing reports
Draw io		It is a free, web-based diagramming tool.

4.6 System Testing and Validation

4.6.1 Connectivity Testing

Connectivity tests were performed between VLANs, servers, switches, routers, and firewall interfaces using ICMP ping operations.
The objective of this test was to verify successful Layer 3 communication across the enterprise network.

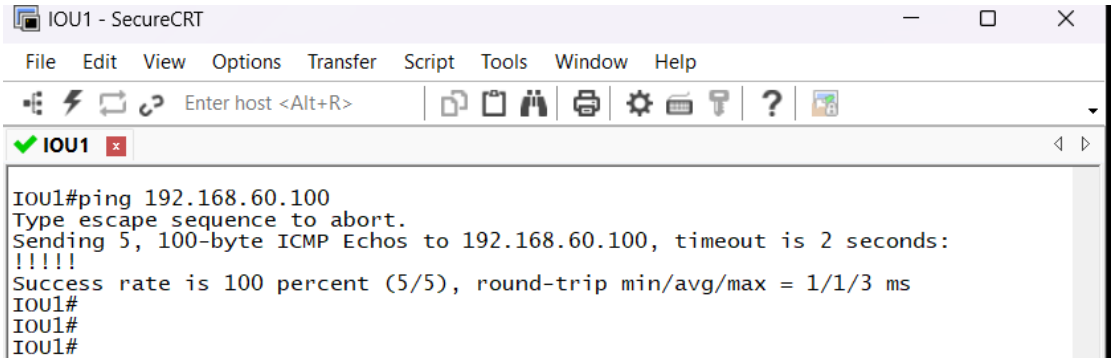


Figure 53:ping form IOU1 to server

Demonstrates successful communication between the multilayer switch and the Windows Server.

4.6.2 VLAN and Inter-VLAN Routing Validation

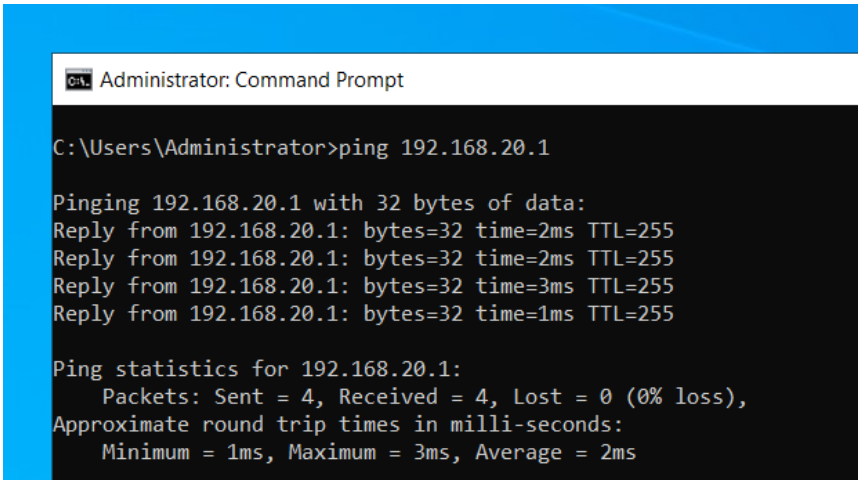
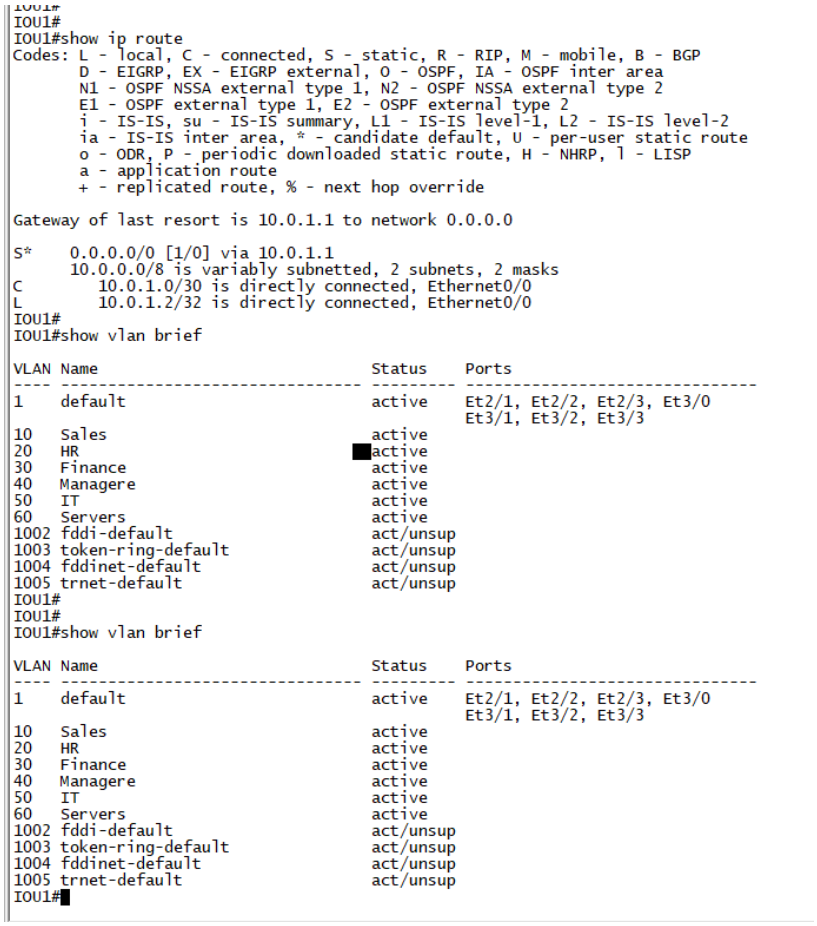


Figure 54:Ping from vlan60 to vlan20



```
IOU1#
IOU1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.1.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.1.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.1.0/30 is directly connected, Ethernet0/0
L        10.0.1.2/32 is directly connected, Ethernet0/0
IOU1#
IOU1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et2/1, Et2/2, Et2/3, Et3/0
                                                Et3/1, Et3/2, Et3/3
10   Sales                            active
20   HR                               active
30   Finance                          active
40   Managere                         active
50   IT                               active
60   Servers                          active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
IOU1#
IOU1#
IOU1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et2/1, Et2/2, Et2/3, Et3/0
                                                Et3/1, Et3/2, Et3/3
10   Sales                            active
20   HR                               active
30   Finance                          active
40   Managere                         active
50   IT                               active
60   Servers                          active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
IOU1#
```

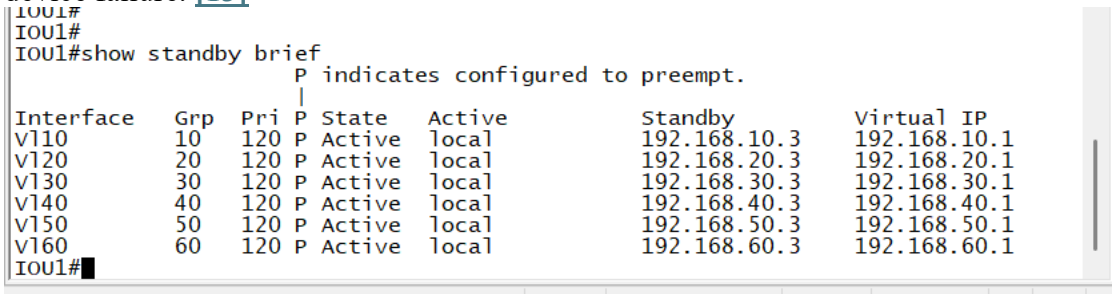
Figure 55:Routing

Routing tables and VLAN membership tables were verified using Cisco IOS verification commands including:

- show vlan brief
- show ip route
- show standby brief

4.6.3 HSRP Redundancy Testing

IOU1 operated as the active gateway while IOU2 remained in standby mode. Failover testing confirmed that network traffic continued successfully after simulated device failure. [13]



```
IOU1#
IOU1#
IOU1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl10        10   120 P Active  local           192.168.10.3    192.168.10.1
Vl20        20   120 P Active  local           192.168.20.3    192.168.20.1
Vl30        30   120 P Active  local           192.168.30.3    192.168.30.1
Vl40        40   120 P Active  local           192.168.40.3    192.168.40.1
Vl50        50   120 P Active  local           192.168.50.3    192.168.50.1
Vl60        60   120 P Active  local           192.168.60.3    192.168.60.1
IOU1#
```

Figure 56:HSRP Active and Standby

```
IOU2#
IOU2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl10        10   100 P Active  local           unknown         192.168.10.1
Vl20        20   100 P Active  local           unknown         192.168.20.1
Vl30        30   100 P Active  local           unknown         192.168.30.1
Vl40        40   100 P Active  local           unknown         192.168.40.1
Vl50        50   100 P Active  local           unknown         192.168.50.1
Vl60        60   100 P Active  local           unknown         192.168.60.1
IOU2#
```

4.6.4 Rapid Spanning Tree Protocol (RSTP) Testing

Rapid Spanning Tree Protocol (RSTP) was tested to prevent switching loops and broadcast storms.
IOU1 was configured as the root bridge to maintain stable forwarding paths across the infrastructure.

IOU1

```
IOU1#
IOU1#
IOU1#show spanning-tree vlan 60

VLAN0060
  Spanning tree enabled protocol rstp
  Root ID     Priority    4156
              Address     aabb.cc00.0200
              This bridge is the root
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    4156   (priority 4096 sys-id-ext 60)
              Address     aabb.cc00.0200
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/1               Desg FWD 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr
Et1/0               Desg FWD 100       128.5    Shr
Et1/1               Desg FWD 100       128.6    Shr
Et1/2               Desg FWD 100       128.7    Shr
Po1                 Desg FWD 56        128.65   Shr


IOU1#
IOU1#
```

Ready Telnet: 192.168.174.128 29, 6 29 Rows, 80 Cols Xterm CAP NUM

Figure 57:RSTP Root Bridge and Forwarding State Verification

```
IOU1#
IOU1#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    4106
             Address     aabb.cc00.0200
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4106   (priority 4096 sys-id-ext 10)
             Address     aabb.cc00.0200
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/1               Desg FWD 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr
Et1/0               Desg FWD 100       128.5    Shr
Et1/1               Desg FWD 100       128.6    Shr
Et1/2               Desg FWD 100       128.7    Shr
Po1                 Desg FWD 56        128.65   Shr


IOU1#
```

4.6.5 EtherChannel Validation

EtherChannel using LACP was configured between the core multilayer switches to provide redundancy and increased bandwidth.

IOU1 IOU2

```
IOU1#
IOU1#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Et1/3(P)    Et2/0(P)

IOU1#
```

Figure 58:EtherChannel and LACP Validation

4.6.6 Firewall Connectivity Testing

```
FortiGate-VM64 #
FortiGate-VM64 # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=119 time=45.5 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=119 time=44.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=119 time=46.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=119 time=46.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=119 time=45.7 ms

--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 44.0/45.7/46.9 ms

FortiGate-VM64 #
```

Figure 59:Internet Connectivity

Connectivity testing was performed from the FortiGate firewall to verify successful internet access and upstream communication.

CLI Console (1)

```
        set service "ALL"
        set logtraffic disable
    next
    edit 7
        set name "LAN-to-WAN"
        set uuid e935c296-48ba-51f1-dadb-8ece304a16f7
        set srcintf "port2"
        set dstintf "port1"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set nat enable
    next
    edit 8
        set name "LAN-to-WAN-FIX"
        set uuid bf59e732-48d2-51f1-2a6a-373904e3a1b0
        set srcintf "port2"
        set dstintf "port1"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set nat enable
    next
```

The FortiGate firewall was tested to verify internal routing communication, NAT functionality, and external internet access.

4.6.7 DHCP Service Validation

DHCP services were tested to ensure automatic IP address assignment for all VLANs. Clients successfully received correct IP addresses, subnet masks, gateways, and DNS configurations.

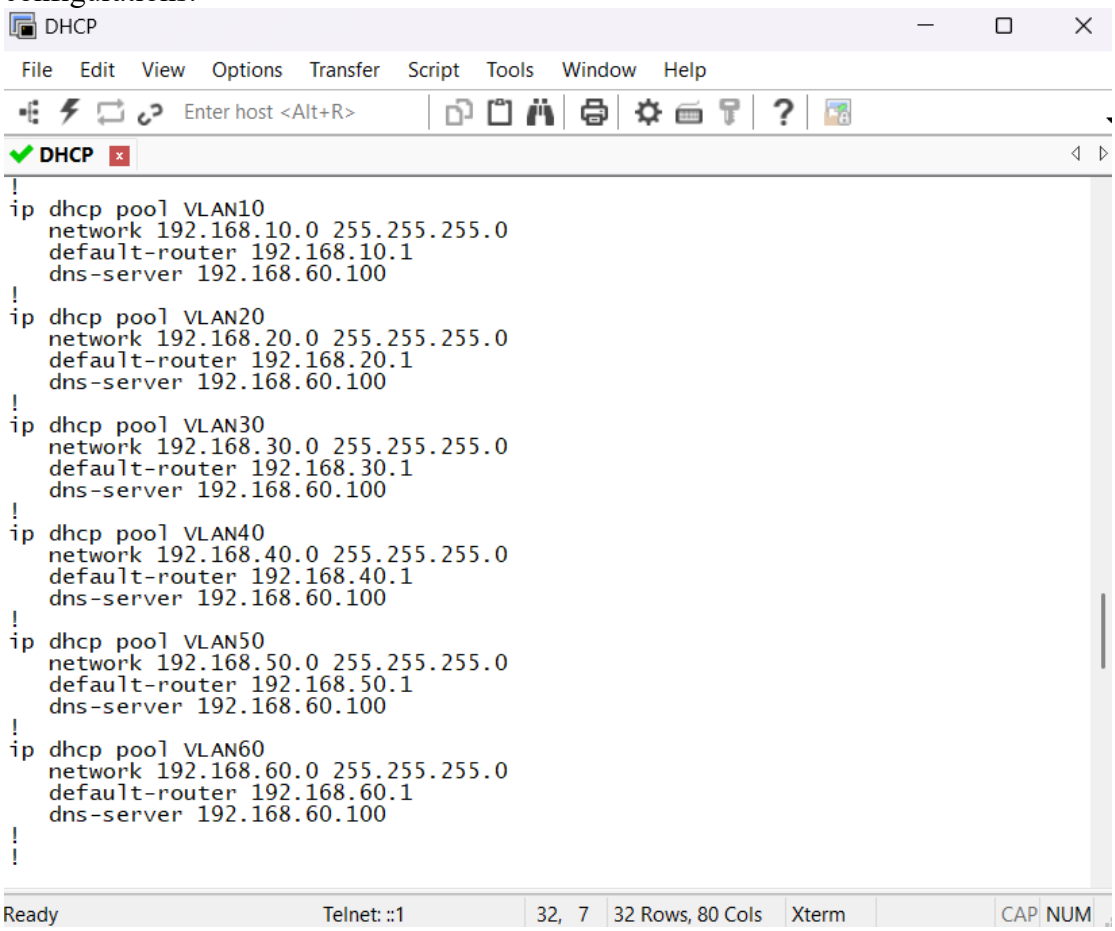


Figure 60:DHCP Pool and Client Address

4.6.8 Dashboard Monitoring Validation

The custom monitoring dashboard was tested to verify real-time infrastructure monitoring capabilities.

The dashboard successfully displayed:

- Online and offline devices
- ICMP latency values
- Active Directory users
- DHCP statistics
- Fault alerts
- Simulated network traffic

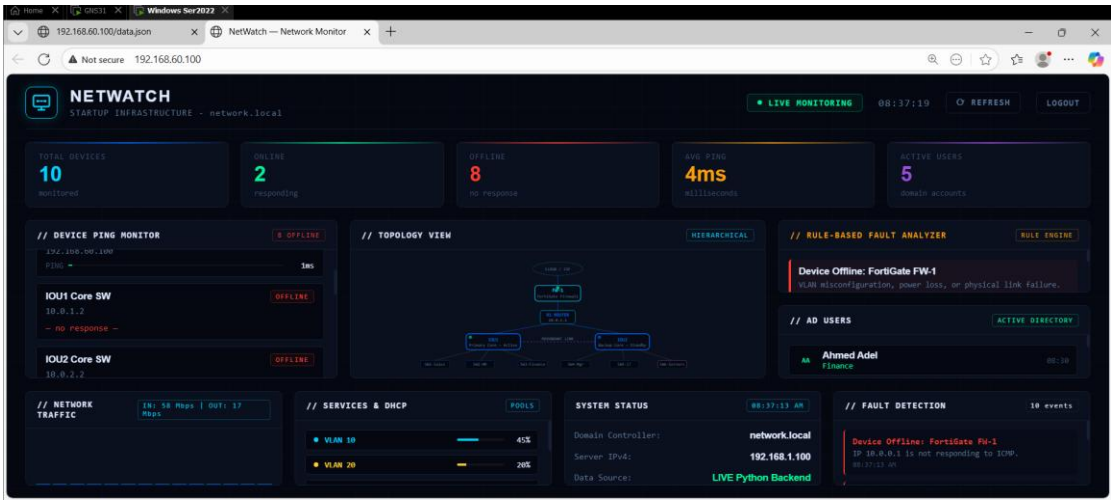


Figure 61:Dashboard



Figure 62:Json

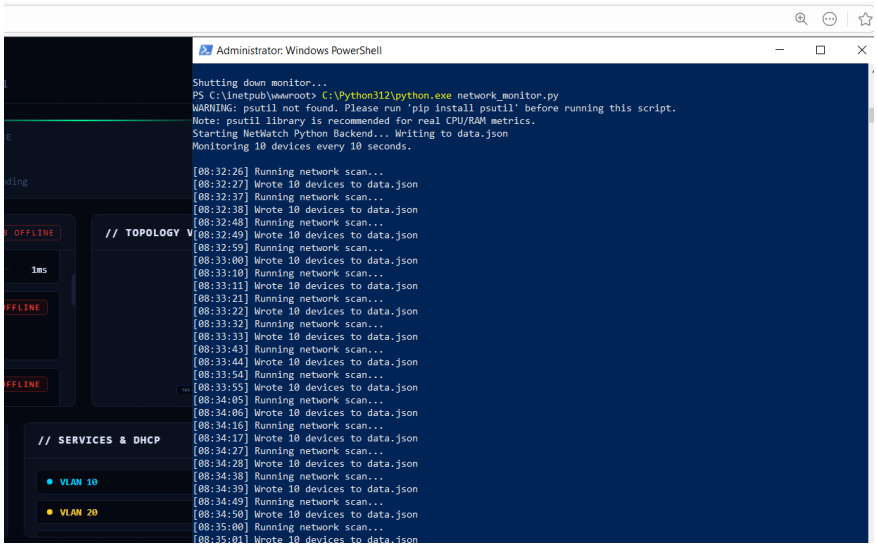


Figure 63:Network scan

Monitoring data was dynamically generated through the Python backend and continuously retrieved by the frontend dashboard using asynchronous JSON requests.

Table 9:Testing

Test Case	Objective	Result
VLAN Connectivity	Ensure devices in separate VLANs can communicate	Communication between VLANs was successfully verified
Inter-VLAN Routing	Confirm Layer 3 routing is operational	Routing tables were verified and inter-VLAN communication functioned correctly
HSRP Failover	Test gateway redundancy	Standby router automatically became active upon primary failure
RSTP Validation	Prevent Layer 2 loops and broadcast storms	Forwarding paths remained stable; network convergence was reliable
EtherChannel	Verify link aggregation between core switches	Logical port-channel remained operational even if a physical link failed
Firewall Connectivity	Confirm external network access	Internal devices successfully reached the internet through FortiGate
DHCP Services	Validate automatic IP assignment for all VLANs	Clients received correct IP addresses, subnet masks, and gateways
Dashboard Monitoring	Test real-time visibility of network devices	Dashboard reflected live device status and metrics dynamically
Fault Detection	Verify automated alerts for device failures	Alerts were generated accurately for offline or unresponsive devices

The testing and validation phase confirmed that the implemented enterprise infrastructure operated successfully under normal and fault conditions.
The integration between routing, switching, firewall security, Active Directory, and real-time monitoring components provided a stable and reliable infrastructure solution.

4.7 Network Security and Protection

The network architecture emphasizes robust security using a layered defense approach. At the perimeter, a FortiGate firewall (FW-1) controls incoming and outgoing traffic to safeguard internal systems from external threats. Inside the network, VLAN segmentation isolates sensitive departments, including Finance and HR, preventing unauthorized lateral access. Group Policy Objects (GPOs) on Windows Server enforce strict endpoint security and manage user permissions to reduce internal risks. The NetWatch dashboard polls the network devices every 10 seconds and raises alerts for unreachable devices and high latency.

4.8 Inclusion, Diversity, and Ethical Compliance

The network design also supports an inclusive and equitable work environment while complying with ethical standards:

- **Accessibility:** GPOs deploy accessibility tools (screen readers, magnifiers, high-contrast themes) automatically to workstations, aiding users with disabilities.
- **Remote Access:** The FortiGate firewall supports IPsec and SSL VPN, which can let employees with health limitations or remote work needs safely access company resources (full VPN deployment is listed as future work).
- **Data Privacy:** Sensitive HR and financial data are isolated in dedicated VLANs with controlled access, adhering to legal and ethical frameworks such as GDPR.

4.9 Resilience and Fault Management

Enterprise networks are prone to hardware failures and link interruptions. To maintain reliability:

- **HSRP Redundancy:** Core switches (IOU1 and IOU2) are configured for HSRP, allowing the standby switch to immediately take over if the primary fails, ensuring uninterrupted connectivity.
- **Proactive Monitoring:** The NetWatch dashboard identifies offline devices in real-time and generates alerts, turning unexpected failures into manageable events for IT administrators.

Summary

This project successfully delivered a **secure, scalable, and fully managed enterprise network** suitable for a startup environment.

Technology Stack: Windows Server 2022, GNS3 simulation, and FortiGate firewall.

Design Approach: Hierarchical network structure with core, distribution, and access layers.

Benefits: Improved network organization, reduced congestion, simplified management, and robust security with continuous monitoring.

Chapter 5

Results and Discussion

5.1 Overview

This chapter highlights the results from implementing and testing the enterprise network infrastructure. It evaluates the system's performance, reliability, and security, and measures how well the solution meets the project objectives.

5.2 Experimental Results

The implementation was highly successful, producing a **fully operational, secure, and monitored network**. A major achievement is the custom **NetWatch dashboard**, which interfaces with the network to provide **real-time monitoring**. Key outcomes observed during testing include:

- **System Uptime and Latency:**
  The dashboard polled the 10 lab devices, showed per-device ICMP latency, and flagged non-responding devices as offline through its fault alerts.
- **Redundancy Validation:**
  HSRP was tested and confirmed effective. **IOU1** serves as the active core switch while **IOU2** remains standby, taking over immediately if the primary fails.
- **Centralized Domain Management:**
  Windows Server 2022 integration was verified. The dashboard lists Active Directory users by department (Finance, HR, IT, Sales, Management), retrieved with PowerShell when run on the domain controller (sample data otherwise); domain login was tested separately (see the table below).
- **DHCP and IP Assignment:**
  Dynamic IP addresses were successfully assigned to end-user devices across all VLANs. This ensured seamless connectivity for every department without manual configuration.

Table 10:Summary of network testing results

Test Case	Description	Result
DHCP	IP assignment to clients	Successful
DNS	Name resolution	Successful
Domain Login	User authentication	Successful
Inter-VLAN	Communication between VLANs	Controlled
Internet Access	Access via firewall	Successful
HSRP Failover	Redundancy test	Successful

5.3 Discussion

he results show that the implemented network is **stable, scalable, and secure**:

- **VLAN Segmentation:** Reduced broadcast traffic and improved security by isolating different departments.
- **Active Directory Integration:** Enabled centralized authentication and simplified management of users and devices.
- **Firewall Protection:** The FortiGate firewall enforces internal and external traffic rules effectively.
- **NetWatch Dashboard:** Provides real-time visualization of device status, latency, user activity, and DHCP information.
- **Fault Analyzer:** The rule-based fault detection system supports troubleshooting by automatically identifying issues and suggesting solutions.

5.4 Goals Achieved

The original objectives of designing a robust, scalable, and forward-looking network infrastructure were largely realized within the simulated environment. The hierarchical model was effectively established, and the
security perimeter was successfully fortified using the FortiGate firewall. Internal segmentation via VLANs was achieved, and the deployment of Active Directory ensured that access control and administrative governance were centralized and strictly enforced.

1. Centralized Management
The implementation of Windows Server enabled full control over users, devices, and permissions. Administrators can now manage the entire network from a single point, improving efficiency and reducing manual errors.

2. Network Segmentation
The use of VLANs successfully separated departments such as HR, Finance, IT, and Sales. This reduced broadcast traffic and prevented unauthorized access between departments.

3. Enhanced Security
Security was implemented using a Defense in Depth strategy, including:
- Firewall filtering using FortiGate
- Access control using Active Directory
- GPO enforcement for endpoint security

This significantly reduced the risk of cyber threats and internal misuse.

4. High Availability
The use of redundancy protocols such as HSRP ensured continuous network operation even in case of device failure, eliminating single points of failure.

5. Real-Time Monitoring

The development of the NetWatch dashboard provided a powerful tool for monitoring:

- Device status
- Network latency
- Active Directory users
- Firewall reachability

This allowed proactive issue detection instead of reactive troubleshooting.

6. Successful Testing

All major functionalities were tested and validated, including:

- DHCP IP assignment
- Domain authentication
- Inter-VLAN communication
- Internet access control
- Policy enforcement

The results confirmed that the system operates efficiently and meets all design requirements.

5.5 Strengths and Limitations.

Strengths:

- The project's main strength is its **layered security (Defense in Depth)** approach and the proactive network management enabled by the **NetWatch dashboard**.
- **HSRP redundancy** provides seamless failover, ensuring high network availability and reliability.

Limitations:

- Being a simulated environment, some physical aspects like **cable quality** or **hardware overheating** could not be fully tested.
- While Active Directory manages users effectively, an **automated off-site backup solution** has not yet been implemented.

5.6 Future Work

- **Cloud Integration:** Move services like backup, DNS, or file storage to platforms such as **Azure** or **AWS** for improved scalability and reliability.
- **VPN Access:** Implement a full secure VPN to allow remote users to connect safely.
- **IPv6 Support:** Upgrade to dual-stack IPv4/IPv6 to prepare for future growth and address exhaustion.
- **Automated Backup and Disaster Recovery:** Ensure data availability and recovery during failures or attacks.
- **Dashboard Enhancements:**
  Convert NetWatch into a full web application
  Integrate **AI-based anomaly detection**



Upgrade the rule-based Fault Analyzer with predictive ML capabilities
Add traffic analysis and performance visualization graphs

Summary

This project successfully designed and deployed a **secure, scalable IT infrastructure** tailored for a startup environment. Using **Windows Server 2022, FortiGate firewall, and GNS3 simulation**, it achieved:

- Centralized management of users and devices
- Strong network security and VLAN isolation
- High availability with HSRP redundancy
- Efficient real-time monitoring through the **NetWatch dashboard**, providing proactive alerts and operational visibility

Overall, the system demonstrates how modern technologies can be integrated to create a **robust, observable, and maintainable network environment**, ready to scale and adapt to future organizational needs.

Chapter 6

Conclusion

6. Final Conclusion

The **"Company System Network Design"** project successfully addressed the multifaceted challenge of designing, implementing, and simulating a complete enterprise-level IT infrastructure tailored for a startup environment. Through this project, a previously unorganized system was transformed into a **secure, scalable, and centrally managed network**, demonstrating how structured planning and modern networking practices can significantly improve operational efficiency.

The network was designed using a **hierarchical model**, separating the system into **core, distribution, and access layers**. This segmentation ensured optimal performance, reduced congestion, simplified troubleshooting, and allowed for easier management of both end-user devices and network resources. The use of **Cisco multilayer switches (IOU1 and IOU2)** enabled **high availability** and **fault tolerance** through **HSRP configuration**, ensuring continuity in case of primary switch failure and eliminating single points of failure. Redundant links, EtherChannel configurations, and Rapid Spanning Tree Protocol (RSTP) further contributed to network resilience and loop prevention.

Security and centralized management were key pillars of the implementation. The deployment of a **FortiGate firewall** provided a robust security perimeter and enforced strict internal and external traffic control policies. VLAN segmentation was applied to isolate departments such as HR, Finance, IT, Sales, and Management, preventing unauthorized lateral movement within the network. This approach enhanced internal security while allowing flexibility in network expansion and policy enforcement.

The introduction of a **Windows Server 2022 environment** played a pivotal role in centralizing administrative control. Integration of **Active Directory Domain Services (AD DS)**, **DNS**, and **Group Policy Objects (GPOs)** ensured consistent authentication, user access management, and policy enforcement across all devices. This centralized system allowed administrators to monitor user activity, manage device permissions efficiently, and enforce security policies systematically across the entire network.

A major contribution of this project is the development of the **NetWatch monitoring dashboard**. This dynamic system transformed the network from a static design into a **near-real-time observable environment**, providing visibility of device status, latency, AD users, and DHCP usage. The dashboard's **rule-based Fault Analyzer** added an intelligent layer to the monitoring system, enabling automated detection of offline devices, high latency issues, and other network anomalies. This feature not only alerts administrators to potential problems but also provides **actionable recommendations**, streamlining troubleshooting and enhancing proactive network management.

The project also incorporated rigorous **testing and validation**, including:

- **Connectivity Testing:** Verification of inter-VLAN communication and Layer 3 routing using ICMP pings and traceroute.

HSRP Failover Testing: Simulation of primary switch failure to ensure seamless automatic takeover by the standby switch.

RSTP and EtherChannel Verification: Validation of loop prevention, port aggregation, and redundancy mechanisms.

Firewall Connectivity and Internet Access: Testing to confirm correct LAN-to-WAN routing and external communication.

DHCP Service Validation: Ensuring accurate IP allocation across all VLANs with correct subnet masks and gateway assignment.

Dashboard Monitoring: Confirming live updates of devices, users, traffic, and alert statuses.

This comprehensive validation confirmed that the infrastructure operates efficiently, maintains **high availability**, and provides **proactive network oversight**, fulfilling all the project's predefined technical, security, and operational objectives.

From an **educational and skill development perspective**, the project successfully combined knowledge from multiple domains, including **network engineering, system administration, cybersecurity, and frontend development**. Hands-on experience with GNS3 simulations, firewall configuration, Active Directory, VLAN management, and Python-based monitoring provided a practical bridge between theory and real-world application. Ethical considerations, such as **data privacy, controlled access, and responsible monitoring**, were embedded into the design, reinforcing best practices in enterprise IT management.

In conclusion, this project presents a **robust, secure, and fully monitored enterprise network solution**. It demonstrates the practical application of hierarchical network design, high availability, security enforcement, centralized administration, and real-time monitoring. By integrating modern technologies such as Cisco multilayer switches, FortiGate firewall, Windows Server, and a Python-based NetWatch dashboard, the infrastructure meets current operational demands while remaining adaptable for future expansion.

The **project bridges the gap between theoretical knowledge and practical implementation**, offering a replicable model for modern enterprise IT environments. It provides a **strong foundation** for scaling, enhancing security, improving management efficiency, and supporting future technological integration, positioning the startup network as a forward-looking, resilient, and observable system capable of sustaining long-term organizational growth.

References

[1] Al-Bahadili, H. (2020) Simulation in computer network design and modeling: Use and analysis. Boca Raton: CRC Press. Available at: https://books.google.com/

[2] Cisco Systems (2023) Enterprise network architecture: Hierarchical design model. Cisco Networking Academy. Available at: https://www.netacad.com/

[3] Fortinet (2023) FortiGate next-generation firewall: Product overview. Fortinet Documentation Center. Available at: https://docs.fortinet.com/

[4] Gibson, D. (2021) GNS3 network simulation guide. Birmingham: Packt Publishing. Available at: https://www.packtpub.com/

[5] Microsoft (2022) DHCP server in Windows Server 2022. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/windows-server/networking/technologies/dhcp/

[6] Microsoft (2023a) Active Directory Domain Services overview. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/

[7] Microsoft (2023b) DNS server role in Windows Server. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/windows-server/networking/dns/

[8] Oppenheimer, P. (2021) Top-down network design. 4th edn. Indianapolis: Cisco Press.

[9] Smith, J. and Jones, L. (2019) 'Implementation of Active Directory services in small business environments', Journal of Information Systems, 35(4), pp. 22–30.

[10] Stallings, W. (2020) Network security essentials: Applications and standards. 6th edn. Boston: Pearson.

[11] Yaseen, N. (2025). From counters to telemetry: A survey of programmable network-wide monitoring. Network, 5(3), 38.
Available at: **https://www.mdpi.com/2673-8732/5/3/38**

[12] Afolalu, O. and Tsoeu, M. S. (2025). Enterprise networking optimization: Challenges, solutions, and technological interventions. Future Internet, 17(4), 133.
Available at: **https://www.mdpi.com/1999-5903/17/4/133**

[13] Syah, T. F. and Adriyanto, S. (2025). Implementation analysis of network core redundancy using HSRP and VRRP protocols for enterprise infrastructure reliability assurance. Journal of Innovations in Computer Science, 4(2), 158–164.
Available at: **https://journal.kawanad.com/index.php/jics/article/view/314**

[14] Rahman, T. and Hidayat, K. I. (2025). Implementation and analysis of Hot Standby Router Protocol (HSRP) for gateway redundancy. Networking Engineering Research Operation, 10(2).
Available at: **https://journal.trunojoyo.ac.id/nero/article/view/31856**

[15] Tonge, A. S., Baniya, B. K. and Gc, D. (2025). Efficient, scalable, and secure network monitoring platform: Self-contained solution for future SMEs. Network, 5(3), 36.
Available at: **https://www.researchgate.net/publication/395396721**
Used in: NetWatch dashboard rationale, **monitoring methodology**, and comparison to existing platforms.

[16] Maasaoui, Z., Merzouki, M. and Battou, A. (2025). A scalable framework for real-time network security traffic analysis and attack detection using machine and deep learning. Platforms, 3(2), 7.
Available at: **https://www.mdpi.com/2813-4176/3/2/7**
Used in: Rule-Based Fault Analyzer concept, **security monitoring**, and traffic analysis in monitoring.

[17] Abdurrahman, A. and Ahmad, B. (2025). Design and evaluation of a fault-tolerant spine-leaf data center network using Cisco Packet Tracer. International Journal of Science for Global Sustainability, 11(3), 334–345.
Available at: **https://fugus-ijsgs.com.ng/index.php/ijsgs/article/view/954**